The Journal of Systems and Software 229 (2025) 112444

Contents lists available at ScienceDirect

The Journal of Systems & Software

journal homepage: www.elsevier.com/locate/jss

In Practice

Organizational factors of software performance testing for systems of systems: A case study using high-reliability organization theory to understand an outage

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ARTICLE INFO

Editor: Dr Daniel Mendez

Keywords:
Software performance engineering
Software scalability engineering
DevOps
Software engineering management
Software security

ABSTRACT

Context: In systems of systems (SoSs), independent constituent systems operated by different organizations cooperate towards a common goal. This empirical paper is the first to explore what these organizations can learn from high-reliability organization (HRO) theory when collaborating on software performance testing, a part of software performance engineering (SPE). Our case study is based on the Norwegian tax report SoS.
Objective: Our overall research objective is to understand the organizational factors of SoS software performance testing. To understand these organizational factors, we also analyze the technical factors of SoS software performance.
Method: Three months before the opening of the tax return system, we observed meetings where three independent organizations coordinated their performance testing. We also attended conference meetings during, and retrospectives after, the opening. These observations were transcribed and deductively coded for the five HRO principles. Further, we analyzed measurement logs from the organizations participating in the tax return system.
Results: The technical root cause of a 90-minute outage in the tax report SoS was a load balancer primary memory shortage caused by too many unserved users. However, the organizational root cause is more interesting: incompatible worldviews reflecting an inadequate holistic SoS management.
Conclusions: HRO theory can benefit organizations preparing for workload peaks in mission-critical SoSs. Using the five HRO principles, the constituent organizations can develop a collective mind, bridging incompatible worldviews in SoS software performance testing.

1. Introduction

On the night of 4 April 2019, the tax reports for 2018 were released in Norway. Four million Norwegians eagerly anticipated checking their eligibility for tax refunds. With this workload peak, the ID portal, the principal Norwegian public authentication system, had an outage of approximately 90 minutes.

While what happened that night in Norway is not unique, it was mentioned in the major Norwegian tabloid and caused stress for the responsible organizations. The ID portal shared resources with the Norwegian common contact register, which was simultaneously unavailable. What would the consequences be if a 90-minute outage occurred during an emergency requiring access to contact information? We use this case of outage to understand and hopefully prevent similar situations.

Three independent organizations collaborated in the Norwegian tax return system. The organizations had independent constituent systems that collaborated towards a common goal as a "system of systems" (SoS) (Nielsen et al., 2015). Such complex systems of systems are increasingly prevalent, but most software performance research has focused on just one component (Han et al., 2023).

The outage in April 2019 occurred despite rigorous technical software performance testing. While performance-related outages have been extensively studied within software performance engineering (SPE), organizational factors have been overlooked by both practitioners and researchers. To understand the organizational factors, we use the theory of high-reliability organizations (HROs) (Weick and Sutcliffe, 2015). HRO theory aims to proactively prevent major incidents within complex mission-critical organizations.

Our overall research objective is to understand the

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https://doi.org/10.1016/j.jss.2025.112444
Received 21 December 2022; Received in revised form 19 March 2025; Accepted 20 March 2025
Available online 21 March 2025

organizational factors of SoS software performance testing. This case study of the 2019 Norwegian tax report system builds on unique access to empirical data exploring the organizational factors involved in software performance testing of an SoS experiencing an outage. Since 2016, we have worked closely with Altinn, the leading organization in this SoS. We started to study the preparations for the 2019 tax return system three months prior to the opening night. We were also present when the outage unfolded during that opening night. Moreover, we attended and analyzed the retrospectives of the 2019 Tax Report SoS.

The *planned* workload peak on the annual tax report opening night challenged software performance. Lessons learned from this planned peak can help organizations better prepare for *unexpected* peaks caused by war, pandemics, terror, or accidents.

We developed four research questions. To achieve the overall research objective, we first analyzed the technical factors that triggered this outage:

RQ1. What are the technical software performance root causes of the outage?

Further, we leveraged HRO theory to understand the organizational causes of the outage. These organizational causes originated during software performance testing; thus, our second question was:

RQ2. What are the organizational root causes of insufficient software performance testing?

Additionally, we extended our inquiry to identify the organizational origins that hindered the effective resolution of software and hardware performance challenges during the outage. We asked:

RQ3. What are the organizational causes of deficient mitigation strategies during the outage?

Finally, to present recommendations for other SoSs facing similar challenges, we sought answers to our fourth research question:

RQ4. What recommendations can assist the management of organizational factors of software performance within a system of systems during testing and operation?

In response to our research questions, this paper provides five contributions derived from applying SOS and HRO theory and principles:

C1. A description of the technical factors related to software performance causing an outage within a system of systems.

As outlined in Section 2.2, papers detailing software performance outages are rare, even more so if we focus on systems of systems. Based on measurement logs and meeting documentation, we have constructed a timeline with relevant parameters, and we describe the technical root causes of the outage. We have also considered the organizational factors behind these technical factors:

C2. A description of the organizational factors related to software performance causing an outage within a system of systems.

This description of organizational factors is based on the transcribed meetings coded for the five HRO principles. To our knowledge, such organizational factors have not been described before.

The three subsequent contributions utilize HRO theory to assist with the software performance of an SoS, first during testing, second during operation, and third by distilling recommendations both for software performance testing and operation:

C3. A demonstration of how high-reliability organization theory can assist in software performance testing of a system of systems.

C4. A demonstration of how high-reliability organization theory can assist with software performance while operating a system of systems.

C5. Recommendations for software performance engineering of a system of systems during testing and operation.

HRO theory has not previously been used to enhance software performance testing or software performance during operation. The corresponding recommendations can be helpful for practitioners and further research.

We start this paper by exploring SoS theory in Section 2, where we also introduce software performance engineering (SPE) and explore HRO. Experiences during previous tax report openings from 2011 to 2018 are examined in Section 3. In Section 4 we describe the organizations involved and the architecture of the 2019 tax report case study. Section 4 also explores the data collection and analysis. In Section 5 we explore the technical causes of the outage in 2019. The five HRO principles describe findings regarding preparations before and experiences during the opening day of the Norwegian 2019 tax report. The findings are discussed in Section 7, along with recommendations. Section 8 outlines what happened in subsequent tax report systems. Factors affecting the study's validity are scrutinized in Section 9, while Section 10 offers conclusions and further work.

2. State of the art

This state-of-the-art section briefly describes research on system of systems (SoS) theory, software performance engineering (SPE), and high-reliability organization (HRO) theory and indicates its relevance to our case study. We detail the five basic principles of HRO theory.

2.1. System of systems (SoS) theory

Nielsen et al. (2015) surveyed the basic concepts of system of systems (SoS) engineering. The inherent tension of SoS is already visible in the first sentence in their abstract: *"The term "System of Systems" (SoS) has been used since the 1950s to describe systems that are composed of independent constituent systems, which act jointly towards a common goal through the synergism between them,"* i.e., the constituent systems are both independent and act towards a common goal. Since the different stakeholders may have *"competing interests and priorities," "an SoS requires trade-offs between the degree of independence in the constituent systems and the interdependence required to reach the common goal."* According to Lopes et al. (2020), there are four levels of coordination between the constituent systems:

1) Directed, where central management has complete authority.
2) Acknowledged, with central management but without complete authority.
3) Collaborated, where the constituents collaborate without central management.
4) Virtual, without both central management and established goals.

By this definition, the 2019 tax report system is an acknowledged SoS.

Nielsen et al. (2015) define SoS engineering as *"a sub-field of systems engineering that focuses on the boundaries and interactions between independent, distributed, and evolving constituent systems and their stakeholders."* The *"discipline of SoS engineering has begun to emerge, aiming to extend systems engineering with the ability to develop, maintain, and adapt SoSs effectively."* The emergent behavior of an SoS adds *"a large degree of unpredictability, as overall system behaviour cannot be predicted by having individual knowledge of each constituent."*

System-level testing of SoSs is required because compositionality may be challenging. Compositionality in SoSs is defined as the ability to *"deduce emergent properties of the complete system from the "local" properties of its constituents"* (Nielsen et al., 2015). Nielsen et al. describe several reasons for violations of SoS compositionality:

1) Some non-functional emergent properties are non-compositional, where combining systems obeying a requirement does not guarantee the same requirement when these systems are combined in an SoS.
2) *"System integration testing on SoS level cannot rely on all constituent systems being ready for this task at a given point in time. The problem becomes even more severe if some constituent systems are members of more than one SoS."*
3) Undocumented system assumptions are not fulfilled in the SoS.
4) Requirements for emergent properties have not been fully captured.

Vargas and Braga (2022) conducted a systematic review of SoS management. SoS management requires close cooperation between systems engineering teams and project management teams. They identified ten SoS management problems, which we also find in our case study:

1) Lack of holistic management view of the SoS.
2) Poor levels of communication and support for interactions between the constituent systems.
3) New flexible mechanisms for project management are needed to cope with emerging dynamic environments.
4) Incompatible worldviews and divergent objectives among multiple stakeholders.
5) Interdependencies among the constituents are not considered.
6) The impact of constituent micro-behaviors on project performance is not considered.
7) SoSs cannot continue to operate using informal approaches.
8) Requirements for common elements may conflict.
9) The constituent may use different terminologies.
10) Traditional project management adopts centralized control, ignoring constituents' autonomy.

After conducting a systematic mapping of risk management in SoSs, Lopes et al. (2020) concluded that there is a tendency to focus on individual risks in each constituent system (CD) without taking a *"holistic view of interactions between the potential risks at the CS level and their consequences to the SoS. In addition, risk management approaches at the CS level do not take into account the influence of external factors and constraints that may affect the overall risks associated with the SoS."*

2.2. Software performance engineering (SPE)

The collection of engineering activities to meet performance requirements is termed software performance engineering (Smith and Williams, 2002; Woodside et al., 2007). Software scalability engineering builds on software performance engineering but focuses more on workload growth (Brataas et al., 2020). Woodside et al. (2007) distinguish between model-based and measurement-based approaches to software performance engineering. According to a survey on industrial practices in software performance engineering, 88 % of the participants in the survey do not apply models (Bezemer, 2019). Our paper documents experiences with a measurement-based approach to software performance engineering.

In a systematic mapping of software performance research, Han et al. (2023) found that only 7 % of the papers were experience papers. Han et al. add: *"Almost all primary studies focus on just one component of a complex multi-component software system for performance enhancement."* In addition to being an experience paper describing a complex multi-component software system, our case study is an SoS.

Our case study parallels previous research in encountering the issue of inadequate specification of performance requirements: *"Very little research examines performance requirements and specifications to detect a performance issue early on. One reason is that for most publicly accessible projects, performance requirements are almost always unavailable"* (Han et al., 2023). Research on bug replication is also rare: *"Research on performance bug replication is almost nonexistent. Bug replication is the first step in the debugging process … still relying purely on … manual efforts"* (Han et al., 2023).

According to Woodside et al. (2007), outage problems caused by inferior software performance are seldom documented: *"By conventional wisdom performance is a serious problem in a significant fraction of projects. It causes delays, cost overruns, failures on deployment, and even abandonment of projects, but such failures are seldom documented."* To our knowledge, no empirical articles have described organizational factors of outages related to software performance in a complex SoS.

2.3. High-reliability organization (HRO) theory

Roberts (1989) initially defined high-reliability organizations (HROs): *"There is a class of organizations that can do catastrophic harm to themselves and a larger public. Within this larger set of potentially harmful organizations, there is a subset which has operated extraordinarily reliably over long periods of time. Operational reliability rivals short-term efficiency as a major goal in these organizations. Extraordinary attention is paid to operational reliability both because of the inherent dangers of the situation and because outcome reliability is impossible to realize without operational reliability. Hence, we call these organizations high reliability organizations."* Roberts presents initial findings for three archetypical HROs: a nuclear aircraft carrier, an air traffic controller, and a nuclear power plant.

There are eight characteristics of pure HROs, according to Roberts and Rousseau (1989): 1) hyper complexity with an extreme variety of components, systems, and levels, 2) tight coupling with time-dependent processes which must be supervised, 3) extreme hierarchical differentiation with multiple levels, each with its elaborate control mechanism, 4) many decision-makers in complex communication networks, 5) degrees of accountability that do not exist in most organizations, 6) high frequency of immediate feedback about decisions, 7) compress time factors where major activities have cycles measured in seconds, and 8) more than one critical outcome that must take place simultaneously.

HRO theory has been applied outside typical HRO organizations, especially in health care (Christianson et al., 2011) and medicine (Sutcliffe, 2011). Cross-organizational systems outside of typical HROs have also been studied using HRO theory. Grabowski et al. (2019) reviewed reliability-seeking virtual organizations in distributed offshore oil and gas systems. Agwu et al. (2019) studied four oil and gas organizations, two beverage manufacturers, and two restaurant chains, and suggested: *"Given that most disasters have occurred in organisations and industries not considered as truly HROs, this paper argues that applying organisational learning from HROs across diverse organisations in different industries could potentially reduce organisational disasters."*

Computer security has benefited from HRO theory. Security is another non-functional requirement like performance and scalability, and there are even interactions between them (Tøndel and Brataas, 2022). The HRO principles reduced risks in the digital operations of F-Secure, a computer security company (Salovaara et al., 2019). While some aspects of software performance testing could be digitalized, manual activities are vital in software performance testing. In the work by Salovaara et al., limitations imposed by the frame problem are essential. The frame problem describes the theoretical impossibility of preparing software for the unexpected: *"An algorithm's capacity to identify and act upon events and states has a fixed upper bound dictated by the current frame manifested in its rules."* The frame problem is relevant in software performance testing because you cannot test something you are unaware of.

The only systematic literature review (SLR) on HRO theory was written by Cantu et al. (2020). In another article, Cantu et al. (2021) looked for tools and techniques for HRO theory and located 26 hits in IT but no hits for performance or scalability testing. Although HRO theory has been applied outside typical HROs for cross-organizational systems and software security, we find no examples of its use in software performance engineering or software scalability engineering during testing or operation prior to this study.

2.4. The five principles of HRO theory

The five principles of HRO theory were initially described by Weick et al. (1999). All these principles are used today, except the wording of the fifth principle has been changed slightly. The textbook that spread interest in HRO theory and the five HRO principles to the broader public was written by Weick and Sutcliffe in 2001 (Weick and Sutcliffe, 2001). The book was refined in its second edition in 2007 (Weick and Sutcliffe, 2007) and its most recent third edition in 2015 (Weick and Sutcliffe, 2015). As it is the most recent and comprehensive textbook on HRO theory, we use this third edition to outline the five HRO principles. All quotes are from this textbook unless otherwise indicated:

- **Preoccupation with failure:** Small, emerging failures may point to future, more dramatic failures. Therefore, instead of hiding failures, *"celebrate them as windows into the health of the system."* (Christianson et al., 2011). *"As a problem unfolds, weak signals are hard to detect but easy to remedy. However, as time passes, this situation tends to reverse as signals become easy to detect but hard to remedy"* (Weick and Sutcliffe, 2015). Weick et al. (1999) wrote, *"HROs are unique because they self-organize to encourage and reward the self-reporting of errors,"* and described HROs as *"collective bonds among suspicious individuals."* Another important insight is, *"Once you accept an anomaly or something less than perfect, it is hard to get back to not accepting it."* HROs anticipate mistakes they don't want to make by asking, *"What do people count on? What do people expect from the things they count on? In what ways can the things people count on fail?"*
- **Reluctance to simplify:** Matching internal and external complexity is core to HRO. *"Simplifications may obscure unwanted, unanticipated, unexplained details. They allow anomalies to accumulate, intuitions to be disregarded, and undesired consequences to grow more serious"* (Weick, 1999). Consequently, the likelihood of eventual surprise and unreliable performance grows. Letting uncertainty build before you simplify may confuse you, but you see more complexity. HROs seek *"divergence in analytical perspectives among members of an organization over theories, models, or causal assumptions pertaining to its technology or production processes… A spirit of contradiction should be organized… This means people are confronted with different points of view, argumentation and criticism are encouraged, controversy is sought and discussed."* Coping with a wide variety of inputs depends on a wide variety of sensors: *"A loan manager who has made good and bad loans is a more complex sensor, able to sense more variety, compared with a loan manager who has made only good loans."*
- **Sensitivity to operations:** This principle is about the work itself, about seeing what we are doing regardless of intentions, designs, and plans. "*Operations are not delegated to some while others think. Instead, HROs think while doing and by doing*." A collective mind is characterized by heedful interrelating where people see their work as a contribution to a system, have a representation of the meshed contributions of others, subordinate themselves to the system, ask what it needs, and satisfy these needs. A serious threat is momentum: "*continuing in a course of action without reevaluation*."
- **Commitment to resilience:** In line with Weick and Sutcliffe (2015), we define *resilience* as *"the capability of a system to maintain its functions and structures in the face of internal and external change and to degrade gracefully when it must."* The overriding principle underpinning HROs *"is not that it is error-free but that errors don't disable it… [HROs] assume that each day will be a bad day and act accordingly… This is a hard state to sustain, particularly if mishaps did not happen, happened a long time ago, or to another organization… High reliability is not a state that an organization can ever fully achieve; rather, it is something the organization seeks or continually aspires to. Reliability is fundamentally a dynamic set of properties, activities, and responses."* Making sense of an emerging pattern can be increased by drilling in advance, investing in the capability to learn and act *without* knowing what to act upon, and recombining fragments of potentially relevant experience. Successful HROs distinguish between three modes of operation: normal, up-tempo, and crisis. All business units should be in the same mode.
- **Deference to expertise:** During an emergency, give decision authority, regardless of rank, to the most highly qualified people *"who can immediately sense the potential problem… This requires loosening the rigid hierarchy, know where the experts are, and have mechanisms to get to them."* Even though the fundamental belief in HROs is that *"sceptics improve reliability,"* good interpersonal skills are essential (Weick, 1999). *"Respectful interaction is the bedrock of shared understanding. … The combination of trust, honesty, and self-respect increases the likelihood that people will speak up about issues of concern, share their perspective and ask others questions about their interpretations"* (Sutcliffe, 2011). It is crucial not to withhold dissent because this reduces the system's ability to detect the unexpected. Therefore, encourage people to reveal information that is not widely shared. Weick and Sutcliffe (2015) add: *"HROs depend on three perishable values: trust, credibility, and attentiveness."*

Cantu et al. (2020) noted that these five principles are essential but *"can be ambiguous and difficult to translate into practice."* Section 6 discusses how these five principles apply to the 2019 tax report case study.

3. Tax report history before 2019

Experiences in previous tax report openings from 2011 until 2018 illustrate the complexity of the solution and show how experiences were gradually accumulated. Performance challenges in the 2015 and 2016 tax reports directly pointed toward the problems in 2019. Reports were written after severe problems, but these reports were not common knowledge among new team members. We first present the overall architecture of the tax report systems before 2019.

3.1. Overall architecture of earlier tax report systems

Before 2019, the tax report system had the overall architecture depicted in Fig. 1. The end users could access the tax report system either (1a) from the Tax front end or (1b) from the Altinn front end. Both these front ends contained a load balancer pair and several web servers.

(2) Users from the Tax front end were redirected to the Altinn front end. (3) The Altinn front end sent the users to the Altinn back end, which (4) redirected the users to the ID portal front end. The Altinn back end had application servers and a database containing the tax reports. The ID portal front end had a load balancer pair and several web servers, and the users were sent to (5) the ID portal back end, which had application servers and a database with authentication information. (6) In the ID portal back end, users selected one of the four authentication methods, where BankID on mobile was the most popular. BankID on mobile redirected users to Telcos, which again had subcontractors. (7) These users were then sent back to the ID portal back end, (8) subsequently back to the ID portal front end, and (9) to the Altinn back end. The Altinn back end finally returned the tax return form to the end users.

3.2. 2011: Official investigation

In 2011, the complete Altinn portal experienced a severe outage for several days caused by performance challenges in its tax report solution. An official investigation concluded that the Altinn web platform had *"limitations and bottlenecks in the architecture that can give the risk that the system is not satisfactorily scalable for future [traffic] volume"* (Hartvigsen et al., 2012). The official investigation proposed that handling capacity problems should be less reactive and more proactive.

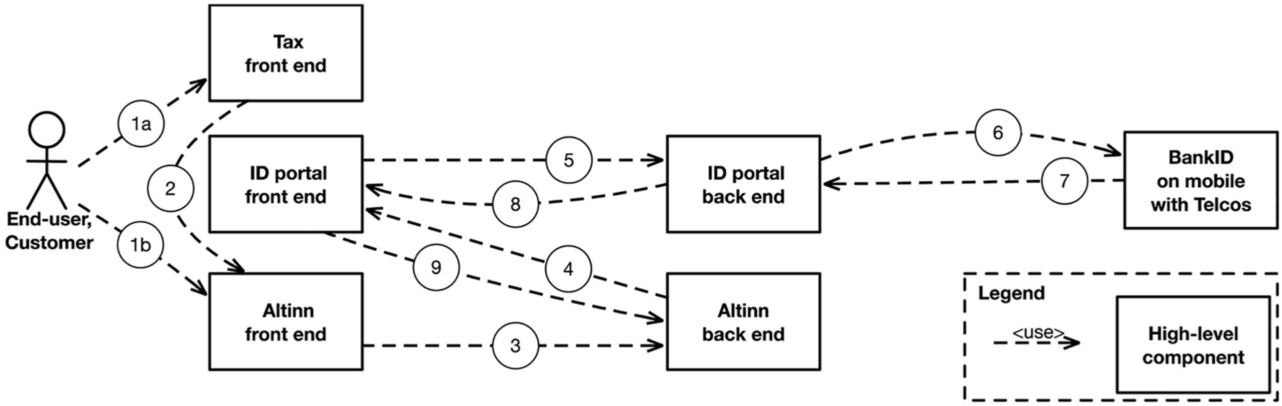

Fig. 1. Overall architecture of earlier tax report systems.

3.3. 2012: Private message box becomes public

In 2012, the tax report system faced new performance challenges. A bug appeared because the Altinn front end was too heavily utilized, delivering dynamic and private cache content as static and public. Consequently, people interested in their tax reports saw another person's message box and social security number. Fortunately, they did not see the other person's tax data. This spectacular bug hit the front page of Norwegian newspapers.

Many years later, the incidents in 2011 and 2012 still caused strong reactions in the Altinn organization. Consequently, data caching was used with great care. Altinn also introduced the costly policy of using <20 % of its hardware resources during high workloads.

3.4. 2013: The Ringelmann effect

The complexity of software scalability engineering was demonstrated in 2013, when Altinn tested its authentication mechanism employing OpenSSO with four hardware servers. Without testing, Altinn installed two more servers just before the tax report opened. However, because of the Ringelmann effect (Brataas et al., 2020), the authentication capacity decreased drastically (Brataas et al., 2017).

3.5. 2014: Educating users and using cache are tricky

In 2014, 900,000 users with an existing electronic ID could read their tax report on 19 March 2014, earlier than usual. However, the remaining 3,100,000 users also wanted to read their tax reports before the regular opening day, 1 April 2014. Consequently, too many users tried in vain to read their non-existing tax reports. Afterward, the users tried registering as electronic users, causing many other public digital systems to go down. After this, the partners were skeptical about educating users about new ways to access the system.

Because of the high workload in the Altinn front end on 19 March 2014, the ID portal was unstable. Therefore, Altinn wanted to publish a notification to end users on their front end. However, since all the servers in Altinn's front end already had high CPU utilization, processing this notification was too much. Additionally, because of a default cache policy, the Altinn front-end servers *had* to refresh their cache content every five minutes from the Altinn back end. This cache policy contributed even more to the already high CPU utilization in the Altinn front-end servers. Consequently, the Altinn front-end servers became slower and slower until none of them responded, and Altinn had a short outage.

Prior to 1 April 2014, the front end had twice the amount of hardware, and the number of front-end servers was increased. The cache refresh in the front-end servers every five minutes was turned off. Altinn's tests showed that increasing the amount of front-end hardware had the desired effect, and the regular tax system opening on 1 April 2014 went smoothly.

3.6. 2015: Routine backup causes problems

In 2015, there were unserved users, just as in 2019. Since the workload is usually low at midnight, routine backups are typically run currently. However, in 2015, DIFI (the organization responsible for the ID portal) forgot to temporarily stop its periodic backup at 00:00. With this large CPU-demanding background job, some components in DIFI got too few hardware resources and low throughput. The ID portal managed to log in users but did not manage to send these users back to Altinn. However, when DIFI eventually found and killed its backup job, the ID portal flushed all its 90,000 logged-in users to Altinn. Consequently, DIFI redirected 1,000 users per second to Altinn. Altinn went down and stayed down until the workload faded.

3.7. 2016: Users refreshing sessions gave a workload flooding

Approximately 60,000 eager users logged into Altinn and refreshed their sessions at 00:00. Simultaneously, the Altinn database made 4,000,000 tax report objects visible. In addition, an average of 200 new users per second and 1,000 new users in spikes tried to log into Altinn. This was too much for the I/O utilization in the TempDB component of the database. Interestingly, the TempDB CPU utilization was moderate at 30 %. As a result of the saturated I/O utilization, Altinn could not serve any users. New users were not logged into Altinn but created over 1,000,000 login requests in a few minutes. After the users went to bed and stopped hammering on Altinn, things went smoothly, and 2016 became a good year with a new record of 2 million read tax reports during the first 24 hours.

3.8. 2017: Telco SMS problem

2017 demonstrated how all parts of the SoS must be alert on the opening day. 2017 was the first year with a silent opening, where the exact opening time at 22:42 was not public information. The official opening was at 00:00, so the flooding problem in 2016 was avoided. Nonetheless, in 2017, the ID portal had problems with Mobile BankID. Whereas the simpler ordinary BankID only uses one Internet call, one Mobile BankID transaction requires four SMSs.

Consequently, one Telco had problems delivering SMSs fast enough. The SMSs sent from Mobile BankID had synchronous connections to the BankID. Because of too many and too long connections, the BankID database ran out of available connections and crashed. A key Telco subcontractor required to solve this issue should have been notified so key personnel were available during the tax return opening.

However, since two other authentication systems, Buypass and MyID (MinID in Norwegian), were working, the complete tax report system was always open for some users. If DIFI had been mandated to disable the two BankID logins, all users could have been directed to the two working authentication systems, Buypass and MyID. Like the year before, the problem disappeared when users went to bed. Thus, except for the breakdown in the two BankIDs, the 2017 tax report system was successful.

3.9. 2018: Mobile BankID had problems

In 2018, the communication between the Mobile BankID and the BankID database was rewritten to use asynchronous connections for one of the two major Norwegian Telcos, while the other Telco did not manage this. As a result, Altinn used a waiting time poster to throttle the load to 73 new users per second. During the 25 April 2019 retrospective, Altinn said that the throttling in 2018 might have masked problems in the ID portal, which surfaced in 2019.

The number of persons using the authentication mechanism Mobile BankID was greater than in 2017. This was too much for the BankID database. As a result, Mobile BankID was disabled in favor of the three remaining authentication systems: (ordinary) BankID, MyID, and Buypass. The disabling of Mobile BankID took more than one hour. Had it been possible to disable Mobile BankID for the Telco with capacity problems, users of the other Telco could still have used Mobile BankID. However, no preparations had been made to disable Mobile BankID for only one Telco.

In 2018, the publicly announced opening of the tax report system was still at 00:00. However, like the year before, there was an earlier silent opening, now at 22:22 in the Altinn front end and at 22:30 in the Tax front end. Except for the trouble with Mobile BankID, 2018 was a successful year for the tax report system. Part of the explanation was the shocking news of a school shooting in the US. As a result, the silent opening was not public information until 22:40. In addition to Altinn's waiting time poster, the workload on the tax return system increased more slowly.

Apart from the BankID, the ID portal managed 180 new users per second during tests in 2018. Altinn managed 500 users per second, while the new cloud-based Tax front end managed 2,000 users per second.

4. Method

This research is based on a case study. According to Yin (2018), case studies are well suited for understanding complex social phenomena as they allow you to focus in-depth on a case and, at the same time, keep a holistic perspective. Furthermore, case studies are well suited to study contemporary phenomena where the researchers have little or no control over events and the boundary between the phenomenon and its context is unclear (Yin, 2018).

In this section, we first explain why the 2019 Norwegian tax report system is an interesting case study for our research questions. We introduce the organizations involved and describe the architecture of the tax report system. Finally, the data collection and analysis are presented.

4.1. Why study the 2019 Norwegian tax report system?

When the first author started working with Altinn in February 2016, he soon realized that the tax report opening day was a considerable performance challenge worth studying. Such a case could reveal insights in line with our research objectives:

1) The workload peak on the yearly tax report opening day led to a high risk of performance challenges.
2) Constituent systems from different public organizations formed a complex SoS.
3) Outages in some parts of the SoS constituting the Norwegian tax report system could severely affect Norwegian society.

Although the data volumes of public digital SoSs may be smaller than those of a private digital system, for example Netflix, they are often more complex. In the public sector, personal, sensitive data must be protected against misuse. Thus, measures to increase performance, such as caching, become complex. (See Section 3.3 for an example of unwanted caching and sharing private data). In contrast, movies on Netflix are easy to cache and share. Moreover, sensitive personal data in public digital systems should be logged, and writing logs complicates performance.

The year 2019 is unique for the tax report system. 2019 was the first year both Altinn and Tax interacted with the ID portal. Moreover, 2019 was the last year with the "Big Bang" approach for the tax report system, where the tax return system was available simultaneously for all Norwegians around 1 April. This workload peak, combined with infrastructure and software modifications, often resulted in unexpected outages (see Section 3). Consequently, software performance testing is crucial. From 2020 onwards, the load was distributed across several days (see Section 8). A distributed load was better for users and the involved organizations. However, as the performance challenges were reduced, the case became less attractive from a performance point of view.

4.2. The organizations involved in the 2019 tax report system

Three principal organizations collaborated in the tax return system. The ID portal ("ID-porten" in Norwegian) was administered by the Agency for Public Management and eGovernment (DIFI). The Norwegian Tax Administration (Tax) handled tax collection systems. Altinn was the leading organization in the tax return system and ran the public Norwegian web portal for electronic dialogue between businesses, citizens, and government organizations.

The cooperation between Altinn, Tax, and DIFI was organized into two teams, as illustrated in Fig. 2. These three principal organizations had subcontractors who assisted with the development of software or with keeping the infrastructure operational. DIFI was responsible for the ID portal, which used one of four authentication methods: BankID, BankID on mobile, Buypass, or MyID. All these four authentication methods used Telcos, which again had further subcontractors.

The *High-Volume Technical Team* included participants from the three principal organizations and several subcontractors. The team members had technical backgrounds and discussed performance and scalability testing from a technical and practical viewpoint. Middle managers from the *High-Volume Steering Team* oversaw the technical team. The steering team was responsible for overall risks and prioritized larger procurements of additional hardware and software. During the preparation from early January until early April 2019, these two teams worked separately but with some contact. However, the steering and technical teams cooperated during the 3–4 April 2019 tax report opening. The manager of both the technical and the steering team came from Altinn. The technical team is termed the systems engineering team in Vargas and Braga (2022), which uses the term project management team for the steering team.

The reason for studying the technical team and not the steering team is a combination of our interests as researchers and the pragmatics of the situation. We were interested in understanding the operational work of performance and scalability testing handled by the technical team. Moreover, the first author already cooperated with Altinn and the technical team manager, who is also the second author of this paper.

4.3. Overall architecture of the 2019 tax report system

The overall architecture of the 2019 tax report system is illustrated in

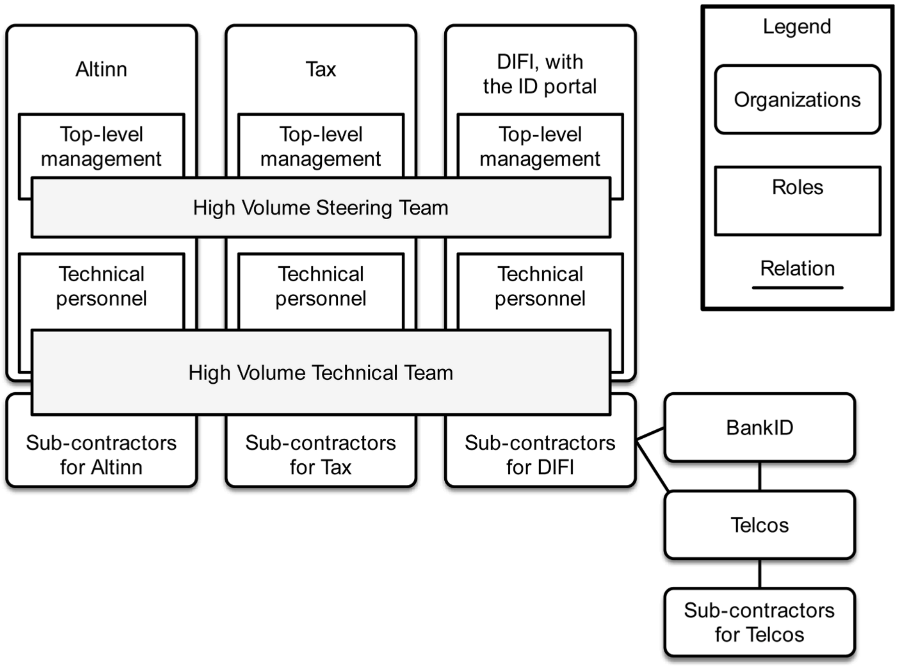

Fig. 2. The organizations involved in the 2019 tax return system.

Fig. 3. This architecture builds on the architecture for the earlier tax report systems, as depicted in Fig. 1. This figure is a simplified version of the official architectural drawings used during the 2019 tax return preparations. However, it also contains essential details that became evident after working with this material in more depth than was possible during the preparations for the 2019 tax return system, i.e., the reauthentication from Altinn in steps 11 and 12 below.

As in 2018, the end users in 2019 had two potential paths into the tax report system: (1a) the Tax front end and (1b) the Altinn front end. However, in 2019, Tax ran a new back end pilot termed Sirius in addition to the existing Altinn back end. Also, the Tax front end was renewed. (2) In 2019, the Altinn front end redirected all its incoming users to the Tax front end. (3) Tax front end sent all users to the new Sirius back end. (4) Sirius redirected all users to the ID portal front end for authentication.

(5) The ID portal front end sent the user to the ID portal back end. (6) Users selected one of the four authentication methods in the ID portal back end. As of 2018, BankID on mobile was the most popular authentication method, and BankID on mobile redirected users to Telcos, which again had subcontractors. (7) These users were then sent back to the ID portal back end, (8) subsequently back to the ID portal front end, and then (9) to Sirius. Sirius handled 10 % of the logged-in users on a “whitelist” and sent the tax return form to the end users.

(10) Sirius redirected the remaining 90 % on the “blacklist” to Altinn’s back end. (11) Altinn’s back end had to verify with the ID portal front end that its users were logged in. This login verification was lighter than a complete login request, but it still consumed storage, processing, and communication resources on the ID portal front end. By storage resources, we mean both primary and secondary storage. Connections are examples of communication resources. (12) After the login verification in the ID portal front end, users were sent back to the Altinn back end again, which delivered the tax report form to the end user.

The new Sirius back end had improved functionality compared to the old Altinn back end. The prime advantage of the Sirius pilot was the dynamic ability to take care of new tax information. In contrast, Altinn used static information from the end of the previous year.

This dynamic feature had no consequences for *reading* tax information on the opening day. However, the capacity of the Sirius back end was lower than Altinn’s back end because of fewer hardware resources and less well-tuned software.

To limit the workload on the opening day, the three principal organizations, Altinn, Tax, and DIFI, had waiting time posters or decline

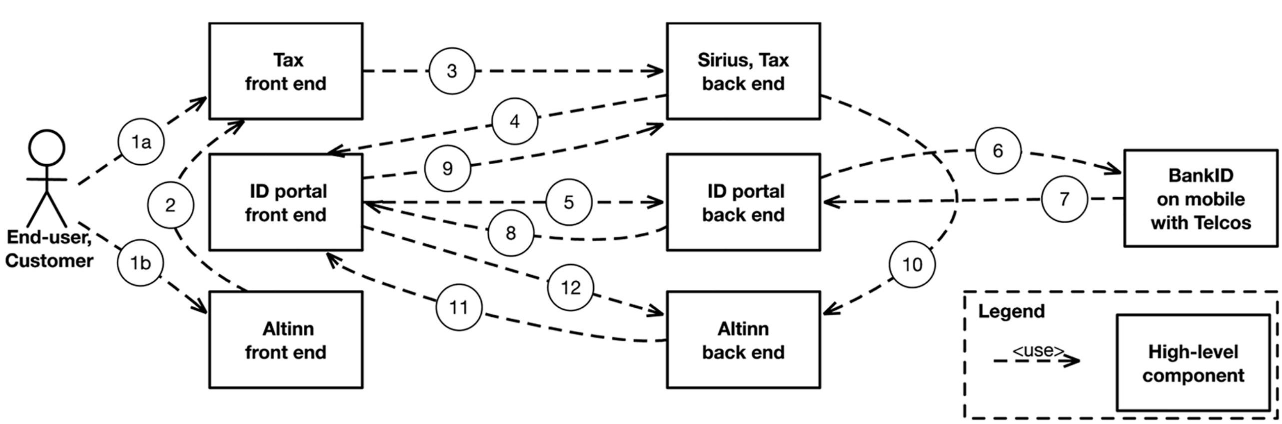

Fig. 3. Overall architecture for the 2019 tax report system.

posters. A *decline poster* is a web page telling users they cannot log in now. The Tax front end redirected an adjustable fraction of users to a decline poster instead of the ID portal front end. The ID portal front end had a decline poster with a maximum load limit. When this limit was reached, users came to a static page telling them they could not log in.

The Altinn front end had a waiting time poster. A *waiting time poster* is a web page informing the user to wait a given amount of time, typically 30 s. If a manually adjustable maximum load limit is not reached, the user gets a waiting poster cookie after this time. Logging into the ID portal front end requires this cookie. However, if the limit is reached, the user must wait again.

As shown in Fig. 3, an unlucky user can first encounter a decline poster at Tax before experiencing a waiting time poster at Altinn and finally meeting a decline poster at the ID portal.

4.4. Data collection

As mentioned in Section 4.1, the research cooperation with Altinn started in 2016, while the tax report system study began in January 2019. The first author followed the team working on the tax report from January 2019 until June 2019. The collected data consists of

- Transcribed observations of synchronization meetings, testing activities, conference calls at the tax report openings on 3–4 April 2019, and retrospectives.
- Written communication in emails and on Slack concerning the 2019 opening day preparations, execution, and retrospectives between January and April 2019.
- Measurement logs from the technical solution on the 3–4 April opening day.

The Gantt diagram in Fig. 4 provides an overview of the data collection before, during, and after the opening day of 2019. This extensive data collection over a long period – before and after the opening day – supports a broad understanding of the situation and how it unfolded, including technical and organizational factors.

See details in the section below.

4.4.1. Observations of meetings

The first author observed the following meetings in the technical team:

- Thirteen weekly synchronization meetings in the technical team from 8 January until 2 April 2019, except on 21 February 2019, during the Norwegian Winter vacation week. These meetings typically lasted 40–60 minutes. The technical teams discussed the status of the performance of the tax report system and planned testing activities.
- From 14 February until 25 March 2019, the three organizations cooperated for 27 h in joint performance testing. In addition, each organization tested the performance of parts of the system independently. In this paper, we focus on joint testing. We had access to the Altinn testing but not the separate testing of Tax and DIFI.

Table 1
Measurement logs.

Agency	Measurements
Altinn	# Altinn users attempting to log into the ID portal
	# unique Altinn users attempting to log into the ID portal
	# logins from Altinn to the ID portal
	# tax reports read from Altinn
Tax	# Tax users attempting to log into the ID portal
	# logins from Tax to the ID portal
	# tax reports read from Tax
DIFI	# logins from either Altinn or Tax with one of the four authentication methods: (ordinary) BankID, BankID on mobile, Buypass, or MyID

- Approximately five hours of conference meetings on the opening day, 3–4 April 2019. One meeting lasted from 21:52 on 3 April 2019 until 01:04 on 4 April and another from 06:19 until 08:06 on 4 April.
- Three retrospectives from 9 April until 25 April with the three main partners. A four-hour retrospective on 25 April 2019. Two 60–75 min retrospectives on 9 and 12 April.

All these meetings were digital, except for the retrospective on 25 April 2019 and the conference meeting on the opening day. On the opening day, the three principal partners had three separate command centers that communicated via video conference. The head of the technical and steering teams worked in Altinn and, therefore, sat in the Altinn team on the opening day.

The first author transcribed recordings from these meetings, except for the retrospective on 12 April, where he only took notes. The first author shared his meeting transcriptions with the technical team, which used them to follow up on previous issues. The transcriptions and the observations were also discussed with the second author, the performance accountant at Altinn, and the head of the technical team. Thus, the second author was also a *participant* in these meetings.

4.4.2. Written documentation

From 1 January 2019 until 30 June 2019, the first author received 324 emails concerning the tax report system. These emails contained meeting invitations, test plans, discussions, and architecture drawings (see figures in Sections 3.1 and 4.3). While writing this paper, the second author (and head of the technical team) filled in details from old emails from the 2019 tax report system that the first author had not received. The first author observed the Slack channel for the tax report system. This channel contained messages before, during, and after the 3–4 April 2019 opening day. The channel also contained material from the tax report system in 2020 and 2021.

4.4.3. Measurement logs

All three partners measured their part of the tax report system. However, these logs were only available after the opening day and not in real time. All log data were directly measured, except for unique attempts where the number of users attempting to log into the ID portal was filtered for unique IP addresses. Table 1 describes the log data in more detail.

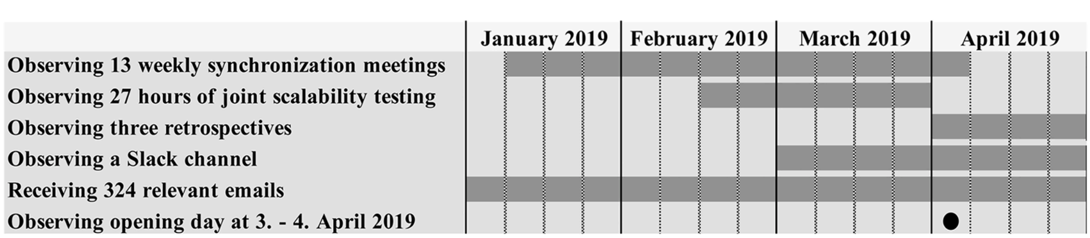

Fig. 4. Overview of the data collection.

4.5. Data analysis

The analysis of the data proceeded along two distinct tracks:

1. The first track focused on investigating the technical causes behind the failure. The first two authors used available documentation from the meetings and measurement logs to identify the critical events and organize them into a timeline. Identifying essential statements from the two video conferences on the 2019 tax report opening on 3–4 April was central to this work.
2. The second track centered on the high-reliability organization (HRO) theory and its application in comprehending the events in this case. This approach facilitated the identification of organizational factors contributing to inadequate software performance testing and subpar mitigation strategies for software performance challenges during system operation. All gathered observations were systematically coded in NVivo, employing a deductive approach based on the principles of HRO, elaborated in Section 4.5.1. The synthesis of these codes corresponding to each HRO principle is detailed in Section 4.5.2.

4.5.1. Deductive coding of HRO principles

We deductively coded the transcribed meetings in NVivo, with the five HRO principles at the highest level. At the next level, we distinguished between what was successful and what could be improved. Altogether, we arrived at 552 codes in NVivo, as further detailed in Table 2. In this table, "level" describes the hierarchical level of a code. "Files" states how many documents the code refers to, and "references" is the number of code occurrences.

The following explanation of the five top-level codes gives more insight into the coding. The top-level code "Preoccupation with failure" was coded 185 times in 24 files and had the following subcodes:

- "Anticipate mistakes you do not want to make" was subcoded into what was done and what could be improved. The former contained only one code on Tax's new Sirius pilot, hitting its workload target during the opening. What could be improved to "anticipate and specify mistakes" had several subcodes for uncovering potential vulnerabilities. One code for uncovering potential vulnerabilities was for lacking risk assessment. Risk assessment was not well integrated into the technical team's work, making it hard for them to anticipate vulnerabilities.
- "Detect small emerging failures" was subcoded into what was successful and what could be improved. Successfully detecting small emerging failures contained several issues, such as making a light web page on the opening day to reduce resource consumption. Improving the detection of small emerging failures focused, for example, on potential bottlenecks in the tax return system.
- "Doubt existing insight" contained improvement issues, such as missing production testing in Tax.
- "Eliminate failure" contained several successful actions, such as preparing the runtime environment for the opening day.

"Reluctance to simplify" was coded 189 times in 23 files with these subcodes:

- "Done to pinpoint variety" with subcodes for representative workload, management of decline and waiting time posters, uncovering capacity before the opening day, adequate capacity during the opening, and discussing the number of test users.
- "Required to see more variety" had subcodes for issues that required more understanding, such as the lack of understanding of the distribution of load between Altinn and Tax.

"Sensitivity to operations" was coded 98 times in 22 files with these subcodes:

- "Done to achieve collective mind" had subcodes for analyzing test results and discussing test scenarios.
- "Improve to achieve collective mind" with subcodes for missing common objectives and missing overview of either the solution or the testing.

"Commitment to resilience" had 74 codes in 11 files with only one subcode for "Improve skills of improvising." This subcode had further subcodes for the missing understanding of what happened on the opening day and for the surprises of poor capacity.

Finally, the fifth top-level code, "Deference to expertise," had only eight codes in four files that focused on symptoms of neglected expertise.

Table 2
Overview of NVivo coding.

Level	Code	Files	References
1	Preoccupation with failure	24	185
1.1	Anticipate mistakes you do not want to make	19	72
1.1.1	Done to anticipate mistakes	1	1
1.1.2	Improve to anticipate mistakes	19	71
1.2	Detect small emerging features	18	57
1.2.1	Done to detect emerging features	12	21
1.2.2	Improve to detect emerging features	9	36
1.3	Doubt existing insights	5	18
1.3.1	Better in the cloud	1	1
1.3.2	Better times for opening	4	10
1.3.3	The ID portal went down too early	1	6
1.3.4	Generally doubting existing insight	1	1
1.4	Eliminate failure	13	38
1.4.1	Improve runtime environment on the opening day	8	10
1.4.2	Sending receipts	6	7
1.4.3	my.policy	5	7
1.4.4	Discussing test priority	2	2
1.4.5	When to start testing	1	1
1.4.6	System owners achieve	5	7
1.4.7	Validate addresses	3	4
2	Reluctance to simplify	24	189
2.1	Done to pinpoint variety	19	111
2.1.1	Discussing the number of test users	6	7
2.1.2	Uncovering capacity before the opening day	8	19
2.1.3	Management of decline and waiting time posters	12	17
2.1.4	Acceptable capacity during the opening	1	5
2.1.5	Representative workload	19	63
2.2	Required to see more variety	13	78
2.2.1	Requirements for the new solution in 2020	2	6
2.2.2	Measures before the next opening day	8	37
2.2.3	A better understanding required	11	35
3	Sensitivity to operations	22	98
3.1	Done to achieve collective mind	15	56
3.1.1	Analyzing test results	11	21
3.1.2	Met analysis needs	5	5
3.1.3	Discussing test scenarios	9	27
3.1.4	Discuss time for joint testing	3	3
3.2	Improve to achieve collective mind	17	41
3.2.1	Common objectives	11	22
3.2.2	Missing information in previous years	6	8
3.2.3	Missing overview of the solution	1	1
3.2.4	Missing overview of the testing	7	10
4	Commitment to resilience	11	74
4.1	Improve skills of improvising	11	74
4.1.1	BankID system and subcontractors	2	4
4.1.2	Workload higher than expected	3	12
4.1.3	Firewall on the ID portal	2	6
4.1.4	Initial theory during the opening	1	1
4.1.5	Missing health information	2	2
4.1.6	Surprises on capacity	4	14
4.1.7	Understanding what happened at the opening	4	35
5	Deference to expertise	4	6
5.1	Symptoms of neglected deference to expertise	4	6
5.1.1	Top-level management in the command center	2	2
5.1.2	The most experienced tester forgets what has been tested	1	3
5.1.3	Facilitate for introverts	1	1

The distinctions between the five principles are blurred, and some of the subcodes could be placed under different principles. Consequently, some subsections were moved between the principles while writing this paper. For example, Sections 6.2.2 and 6.2.5 were initially located in the "preoccupation with failure" section since they describe potential failure. After closer examination, it was more striking that aspects "disappeared" after simplification than missing preoccupation with potential failures. Therefore, this fits better under the "reluctance to simplify" principle.

4.5.2. *Synthesizing HRO codes to find organizational causes*

After coding the files, we analyzed and synthesized codes that corresponded to each High-Reliability Organization (HRO) principle in Section 6. The code synthesis provided us with subthemes for the HRO principles. The five principles and the relevant subthemes are presented in Section 6, HRO Findings in the 2019 Tax Report System.

Section 7 prioritizes these organizational causes based on their impact on the outage. It employs the ten common SoS management issues outlined in Section 2.1 and formulates recommendations for enhancing future software performance testing within SoSs.

Assisted by the other authors, our joint insight into the problem has matured since the 2019 tax report opening. Gradually, we utilized more data sources. For example, in the first version of the recommendations sent to Altinn on 13 January 2020, the first author said that the technical teams should capture risks more systematically *"by looking at a detailed architectural sketch and mapping the risks and consequences for different parts of the architecture."* When discussing drafts with the second author, it became clear that the steering team had done a risk assessment around 1 March 2019 without involving the first author. This risk assessment in the steering team is further described in Sections 6.3.2 and 7.1.3.

5. What happened at the 2019 tax report system opening?

This section provides an overview of the 2019 tax report system opening. Overall, the opening on 3–4 April 2019 was successful. The 90-minute outage is elaborated in a timeline, after which the eleven phases of the outage are analyzed. Finally, the software performance technical root cause of the outage is described in Section 5.4, answering RQ1.

5.1. *Successful overall outcome*

The 2019 tax report system was successful overall. Tax noted on 9 April 2019: *"We are mostly satisfied. Despite the problems with the ID portal, it does not look like we lost so many users. The figures show that those most curious could calm their curiosity at about the same time as last year."* The organization responsible for the ID portal, DIFI, said in the retrospective on 25 April 2019, *"Now we have discussed [what happened at] night, but the whole next day, the traffic flowed smoothly."*

Officially, the tax return system opened at 00:00 on 4 April 2019. In practice, it opened at 22:18 in a "silent" opening. A silent opening means that the exact opening time was not public information. The silent opening in 2019 was introduced because of experiences in 2016 (see Section 3.7). From 22:18 on 3 April 2019 until 24:00 on 4 April 2019, 2,344,737 persons managed to read 3.26 million tax reports. Except for a 3.14 % higher number in 2018, this was a new record for reading tax reports during the first planned 24 h. 63.4 % of the tax return requests originated from Altinn, and the rest from Tax.

2019 was the first year Tax planned to handle a fraction of the workload with its Sirius pilot. During the retrospective on 25 April 2019, Tax *"was satisfied with the test of the Sirius pilot, which got more than the projected 7 % of the load. It got between 10 and 11 % of the users."*

5.2. *Timeline of outage*

This section illustrates the 90-minute outage in a three-hour timeline. As outlined in Section 4.4.3, the timeline in Fig. 5 primarily relies on log data from Altinn and Tax, with less input from DIFI. The information in Fig. 5 was unavailable at runtime for the three main partners observing the tax report opening, but was compiled by the first and second authors.

Three curves in Fig. 5 depict *logins* to the ID portal (front and back end) per minute from Tax, Altinn, and combined. These three curves were based on logs from the ID portal with logins from Altinn, Tax, and combined. The four other curves come from Altinn's waiting time poster and Tax's decline poster.

As explained in Section 5.1, there was a silent opening at 22:18, before the official tax system opening at 00:00. Around 22:22, tweets from the public communicated that the tax report system was open (and no longer silent). As a result, the combined logins (from both Tax and Altinn) to the ID portal passed 3600 per minute at 22:23. At 22:36, DIFI communicated in the coordination meeting that they were *"close to decline poster [limit],"* i.e., they operated at close to maximum capacity. At 22:43, VG, the largest tabloid in Norway, reported that the tax report was available. The ID portal operated at full capacity of 5,000 logins per minute at 22:46.

At 22:49, the ID portal had an outage with virtually no ID portal system until 23:06. From 23:06 until 23:19, the ID portal was almost up before it went down again. In Altinn, putting the waiting time poster to zero at 23:30 had not been tested. From Fig. 5, we see that the load from Altinn *increased* after 23:30. Even if Tax was considerably slower in limiting the traffic to the ID portal, Fig. 5 shows that Tax is finally more successful than Altinn after 23:32. At 00:05, the ID portal worked as expected for Altinn users, and at 00:19, it worked for Tax after being essentially unavailable for 90 minutes.

Altinn attempts are Altinn users attempting to log into the ID portal, *with or without success* (see Table 1 for details). These users have passed the waiting time poster in the Altinn front end and have received a waiting time poster cookie. The waiting poster cookie allows login to the ID portal. Users with the waiting poster cookie can ask for a login to the ID portal and are then redirected from the Altinn front end to the ID portal. If the ID portal does not log them in, they can repeatedly log in to the ID portal. A web cookie is typically valid for 30 minutes.

Similarly, *Tax attempts* are Tax users attempting to log into the ID portal. These users have passed the Tax decline poster and received a decline poster cookie from Tax.

Unique attempts are login attempts coming from unique IP addresses. Residents who live in the same apartment buildings may share the same IP addresses, so the number of login attempts may be too conservative, meaning there are more unique attempts than this curve tells. Tax had trouble filtering for IP addresses. Therefore, unique attempts from Altinn were multiplied by a factor representing the fraction of ID portal logins from Tax to estimate the total unique attempts.

Fig. 5 shows significant unserved users from both Tax and Altinn. The ID portal did not serve these users, indicating substantial problems with the decline and waiting time posters of Tax and Altinn. In Fig. 5, these unserved users are termed "Tax attempts" and "Altinn attempts." These users have *passed* the decline and waiting time posters. The whole point of these posters was to limit the users, but when it mattered, they did not work.

The *accumulated queue* is the difference between unique login attempts and total logins. This number is calculated based on log data. During the outage, the number of unserved users, primarily from Tax, increased drastically, as illustrated by the accumulated queue of unserved users surging just before 22:49. This accumulated login queue continued to grow until it reached 465,000 users at 00:16. From 23:43 until our period ends at 01:00, >400,000 users are in the accumulated login queue.

An investigation by the operations subcontractor of DIFI, dated 10 April 2019, found that between 22:18 and 23:18, the ID portal showed 5 million decline time posters, most of them in the last 30 minutes. Assuming 3.6 million decline time posters between 22:48 and 23:18 translates to an *average* rate of 120,000 ID portal login attempts per

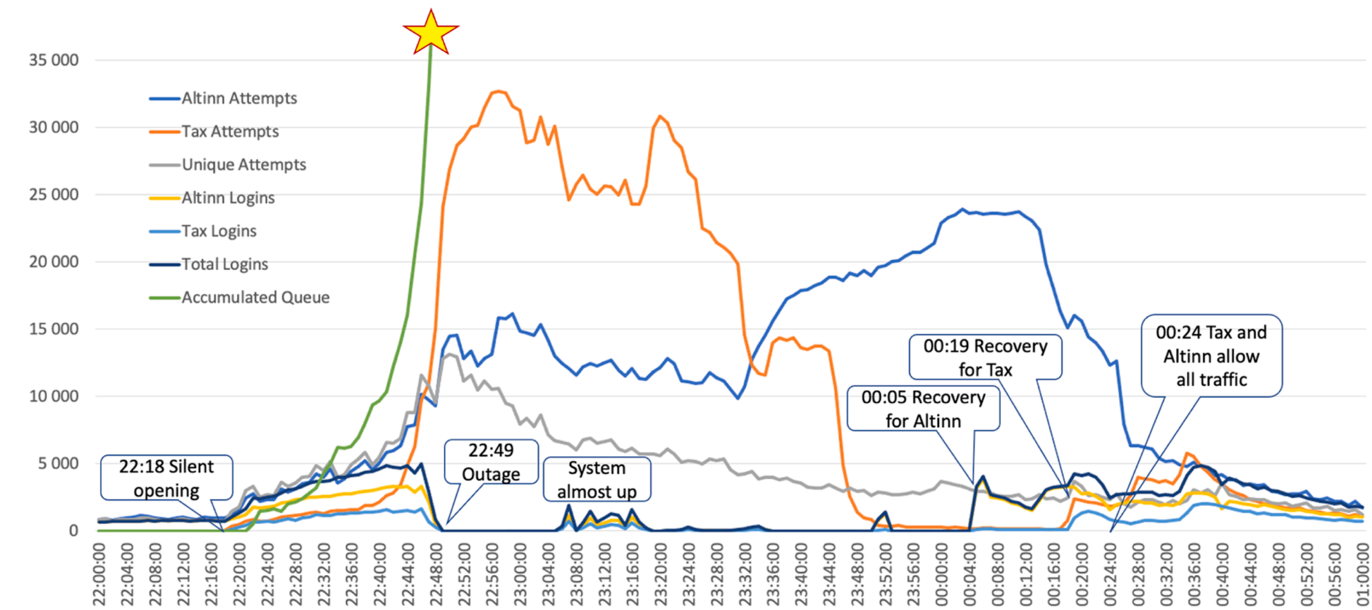

Fig. 5. Timeline for the tax report on 3 and 4 April 2019, illustrating logins to the ID portal per minute.

minute or 2,000 attempts per second, considerably more than the number of attempts from Altinn and Tax, as shown in Fig. 5. The first time a user enters either the Altinn or Tax front end, the user gets a cookie from Altinn or Tax posters and is counted by either Altinn or Tax. However, if the user tries again, the user goes directly to the ID portal front end and is not counted by the Altinn or Tax posters. As the ID portal went down, even more users should have received a decline time poster from the ID portal. Moreover, Altinn users already logged into the ID portal by Tax required an extra lightweight login verification by Altinn (see Section 4.3). Since this is neither counted as a login nor a login attempt to the ID portal, this is not shown in Fig. 5.

DIFI was responsible for the common contact register, which was also unavailable because of the outage in the load balancer. For a potential emergency requiring access to contact information for sending SMSs or emails to many persons, an outage of 90 minutes could lead to severe consequences. It also impacted the ability of Norwegian citizens to authenticate for a broad range of public and private systems.

5.3. Eleven phases of the outage

This section distinguishes between different phases of the outage. Table 3 below presents the eleven phases after the silent opening at 22:18 on 3 April 2019 until 01:00 on 4 April. This table offers a different view of the timeline in Fig. 5. Each phase in Table 3 has the following columns:

- Phase number
- Description
- Start time
- End time
- The average number of login attempts per second in each phase
- The average unique number of login attempts per second in each phase
- The average number of ID portal logins per second

See Section 5.2 for a detailed description of the measurements.

In the bottom row in Table 3, we see that the unique number of attempts for the whole period, on average, gives 76 unique login attempts per second for the total period of 2 hours and 42 minutes. This is slightly less than the ID portal's expected capacity, around 83 logins per second (or 5,000 logins per minute). Theoretically, with a resilient solution assisted by resilient organizations, it should be possible to serve all interested users before 01:00 on 4 April.

If we acknowledge that the average number of unique attempts per second is lower than 76 in the first phase, it should still have been possible to serve all users. On average, there are 78 unique attempts per second from phase two through phase eleven. Nevertheless, the number of unique attempts in some phases is too high for the ID portal to handle, with a top of 166 unique attempts per second in the seven-minute phase 4. Therefore, some waiting will be required.

During the successful ramp-up in phase 1, we see a good balance between unique login attempts and the number of persons logging in. Especially in phase 4, there are many attempts from unserved users, with a seven-minute average of 735 attempts per second. However, in the final phases, 10 and 11, the number of persons logged in exceeds the

Table 3
Eleven phases of the tax report 2019 outage.

#	Description	Start time	End time	*All attempts per sec*	*Unique attempts per sec*	*ID portal logins per sec*
1	Successful ramp up	22:18:00	22:36:59	66	56	47
2	Troubled ramp up	22:37:00	22:46:59	168	121	77
3	On the way down	22:47:00	22:48:59	375	161	31
4	Outage, complete	22:49:00	23:05:59	735	166	0
5	Almost recovered	23:06:00	23:18:59	624	117	15
6	Outage, almost	23:19:00	23:35:59	568	90	2
7	Outage, complete	23:36:00	23:50:59	465	62	0
8	Almost recovered	23:51:00	23:52:59	334	47	19
9	Reset	23:53:00	00:04:59	368	52	0
10	Altinn gives traffic	00:05:00	00:18:59	365	43	46
11	Regular operation	00:19:00	00:59:59	139	38	52
	Total period	**22:18:00**	**00:59:59**	**351**	**76**	**29**

number of unique login attempts.

5.4. Technical root cause of the outage

Testing in January 2020 documented that too many users exhausted the primary memory in DIFI's load balancers. DIFI's load balancers also handled DIFI's decline poster, i.e., the load balancers and the decline poster shared hardware resources. Section 5.3 describes a seven-minute average of 735 attempts per second, measured by Altinn and Tax. A DIFI investigation discussed in Section 5.2 documents a 30-minute average of 2,000 ID portal login attempts per second. Later, in Section 6.2.2, we describe users that were not even counted: unserved users receiving time-outs instead of an ID portal decline poster. Altogether, this heavy workload was too much for the load balancers.

Several factors consumed resources in DIFI's load balancers:

- The primary memory was exhausted because of all the unserved users. Each unserved user consumes several connections, each of which consumes a small amount of primary memory.
- Swapping memory to and from disks to compensate for the primary memory shortage exhausted the processors in the load balancers.
- Since users stayed unserved in the load balancer for a long time, the load balancer used more connections. Consequently, the load balancer also went out of licensed connections.

Because of the incident in the load balancers, DIFI's firewall also ran out of connections. While the firewall's lack of connections was spotted, the problems in the load balancers went unnoticed.

In theory, autoscaling in the ID portal could have prevented this outage. However, autoscaling was not applicable since the ID portal had on-premises hardware, where some resources were fully utilized. Even if autoscaling had been a viable option, exhausting the load balancer's primary memory happened so fast that the load balancers did not notice. Therefore, the load balancers could not even notify their own system logs. Without detection, notifying an auto-scaler is impossible. Section 8 elaborates on how even a cloud-based system with autoscaling encounters problems when users arrive too fast.

Three theories regarding the root cause of the outage were formulated on the opening day on 3 April 2019 (see Section 6.4.3 for details):

- The first theory was presented at 22:51 when DIFI, the organization running the ID portal, discovered (at 22:48) the outage and suspected it was due to a Mobile BankID error.
- The second theory was at 22:55 when the ID portal denied access to foreign IP addresses. DIFI thought they might be victims of a DDoS attack.
- The third theory was at 23:54 when Altinn asked if DIFI ran out of connections on the load balancer, and DIFI responded: *"It may be the firewall that is running out of connections, not the load balancer."*

The third theory was held until the testing in January 2020. Testing in January 2020 documented that too many users exhausted the primary memory in DIFI's load balancers.

We have now explained the *technical* root cause of the outage. In Section 6 we investigate the *organizational* causes of the outage using HRO principles as our framework. Section 7 summarizes the findings and provides recommendations.

6. HRO findings in the 2019 tax report system

This section links the principles of HRO theory with the findings of the Norwegian 2019 tax report case study. The five HRO principles include relevant subthemes that became apparent during the analysis and synthesis of the coded data. For each of the five HRO principles, we present our findings from experiences and situations observed during the time leading up to, and during the day the tax report system was opened for users. To preserve anonymity, we present statements from different partners' names, e.g., "Altinn said," instead of using an individual's name.

6.1. Preoccupation with failure

In line with the "preoccupation with failure" principle, the technical team exploited experiences gained from failures during previous tax report openings (see Section 3). The technical team streamlined web pages, making them easier to retrieve on the opening day, and tested the capability to broadcast messages to users about potential problems. The required amount of processing, storage, and communication resources was estimated based on performance testing.

Our findings showed that being preoccupied with failure can be in tension with efficiency. For example, the time needed to test all potential bottlenecks of the complex tax report system may be insufficient. Several apparently resolved issues that were intended to avoid failure revealed hidden complexities:

- The Tax front-end's load balancers employed a global caching policy to manage data storage between high-speed memory and disk storage. In the meeting on 12 March, the technical team discussed how three-second delays in the global policy likely triggered time-outs. Although initial testing with Tax's Gatling load generator used only four IP addresses, later tests revealed that these timeouts would not occur with unique IP addresses. Consequently, the team removed the global policy from the test priority list. However, since users in housing associations might share the same IP address, this global policy problem may have contributed to login delays.
- Global logout means that if one system logs a user out, the other systems must agree that system is logged out. In earlier synchronization meetings, some thought global logout did not impact performance. Just before the opening day, global logout worked, but it was a heavy call. However, this heavy call was only relevant to the relatively few users *inside* the solution.
- With Tax's smart button, the browser was responsible for throttling the workload, which seemed wise since the browser was the first component in the SoS. With 10 % increments, Tax front-end users could either login to the ID portal or display a high-traffic message. Tax's smart button was extensively tested with a synthetic workload and worked for *new* users. On the opening day, it became apparent that *existing* users had to refresh their web pages *manually*. Otherwise, the *automatic* web page refresh frequency was 30 minutes. Consequently, the response was slow both when allowing fewer users (after 22:49) and allowing more users (after 00:05).

The subtheme of doubting insights into other teams was visible in the synchronization meetings:

- In the first synchronization meeting on 8 January, Altinn questioned missing production testing of Sirius, but Tax did not change this decision. Tax employees were instructed to test their own tax reports to partially compensate for this risk.
- On 12 March, Altinn said: *"In summary, the only traffic restriction planned so far is the CPU quota on the load balancer."* Tax replied: *"That's right."* Altinn suggested using the smart button to protect not only the ID portal but also the Sirius back end.

As doubting insights in your own team typically happens *outside* the synchronization meetings, this was not a part of our study.

6.2. Reluctance to simplify

The HRO principle of reluctance to simplify aims to ensure that details and perspectives are not overlooked. However, our findings show that simplification had negative consequences.

6.2.1. A representative workload

Effective performance testing requires a synthetic workload that accurately mirrors real-world usage patterns. If the test workload is oversimplified, it will not represent essential aspects of actual user behavior. The organizations regularly discussed workload parameters in their synchronization meetings to ensure their tests would reveal potential performance issues before the system went live. The technical team had control over some workload parameters early, for example, the size of the tax reports, with an average size of 71.5 KB and a maximum size of about 120 KB. Differences in test tools between Tax and Altinn amplified the workload discussions. Tax used the Gatling performance testing tool, while Altinn used Visual Studio. Think times were handled differently in the two test tools. The think time is the time between receiving the result of one request from the (computer) system and sending a new request (typically by a human to the computer system). Tax estimated a think time of 30 s, while Altinn ran without think times.

On 29 January 2019, Altinn asked if Tax, with Gatling, accessed images and style sheets on the web page during testing, like in Visual Studio in Altinn. There was no immediate reply from Tax. Also, on 28 February, they discussed differences in the test tools. On 5 March, they agreed not to prioritize this difference between the test tools. Immediately before the opening day, on 2 April, Altinn said: *"What we gave up was to get similar test scripts, but it is not certain that it has been a disadvantage to run double scripts in some solutions."* In the same meeting, Tax said: *"It's OK with two different verifications."*

Since Tax, in contrast to Altinn, ran with think times, the same arrival rate resulted in more users with more connections. It also resulted in memory usage in the load balancers since each connection used a small amount of memory. This may explain why Tax discovered problems in the ID portal during testing, as described in Section 6.2.2. In line with the "reluctance to simplify" principle, a more complex understanding with divergent analytical perspectives in two test tools gives more insight, but adds complexity.

On the opening day, the distribution of the four login methods was 43 % BankID, 41 % Mobile BankID, 15 % MyID, and 1 % Buypass. During testing, Altinn and Tax used the login method MyID, which was easiest to simulate. They compensated by running with a higher workload, which may camouflage the side effects of using the heavier (ordinary) BankID, especially Mobile BankID. However, a heavier call held connections longer.

There were also other aspects of the workload which were simplified. In the 25 April 2019 retrospective, the technical team discussed that Mobile BankID polls the ID portal once every second to see if it has an authentication. However, this was not part of the performance testing of the ID portal. In the retrospective, the technical team also discussed other surprises. Altinn said: *«We had an hour where one [user] was logged in 116,000 times."* The organization responsible for the ID portal, DIFI, replied: *"The user may have been in a loop situation. We see in our graphs that someone gets into a loop between Altinn and us."*

With a deeper understanding of discrepancies in synthetic workload among the partners, the technical team would have understood that running with think times for a sufficiently long duration would force the ID portal to go down and would have done something about this. Moreover, if also Tax had loaded more graphics, the testing would have been even more demanding, and the ID portal might have gone down earlier.

6.2.2. Outage root cause almost found

During the tax report preparations, a simplified interpretation of test results failed to reveal the technical root cause of DIFI's load balancers running out of memory, causing the ID portal to fail (detailed in Section 5). While aspects of this technical root cause were discussed in several synchronization meetings, simplification obscured unwanted, unanticipated, unexplained details, allowing undesired consequences to grow more serious (Weick, 1999).

According to Tax in the retrospective on 25 April 2019: *"We had an episode with our tests where we were told to stop, and we received a report that we used up connections, threads, and the load balancer's capacity. Was it the same thing that happened in production?"* This was mentioned in the synchronization meeting on 5 February, where Tax said: *"As you know, when we ran hard from our side, we dragged down the PT1 [Performance Test 1] environment at the ID portal at least twice. We have not fully understood why and have thought we should delve into this, but as I understood from the meeting on [last] Friday, the [decline] poster [of the ID portal] has not been tuned properly."* This quote illustrates Tax's "preoccupation with failure" and its willingness to investigate the problem in the ID portal. The quote also illustrates holistic SoS management.

However, this issue was more complex. Immediately in the same meeting, Tax added: *"On the test last Friday, we had the supplier [our subcontractor] with us, and we ended up with 500 logins per second and redirected 420 [logins] to the ID portal per second. We are losing some [of the logins], so we need to investigate this. Then it [the decline poster of the ID portal] was probably tuned so that it allowed 100 logins per second."* DIFI, the ID portal organization, concludes later in the meeting: *"We have shown that we manage 4 - 500 logins per second well. It was a good test for us too"*.

DIFI concludes that these performance tests were successful, but this simplification ignores two things: First, while the latter test was successful, this was not the case for the two earlier tests referred to in the same meeting. The cause of this discrepancy in outcome between the tests may be the length of the test. Tests for a longer time than the standard 10 minutes would give more connections for the ID portal and, therefore, more likely result in the same problems as on the opening day. Second, 80 ID portal logins per second were lost in this last test, and the technical team did not know why. In retrospect, missing logins may be time-outs because of missing responses from the ID portal, which became apparent later.

Implicitly, the technical team assumes only two types of users: 1) Users logged into the ID portal. 2) Unserved users who received a decline poster from the ID portal. The ID portal counted both these types of users. However, a third type of user was unserved users receiving time-outs instead of an ID portal decline poster because of heavily utilized resources. These heavily utilized resources eventually resulted in the ID portal outage. The ID portal did not count this third group of users during performance testing, and these users were not included in the investigation by DIFI's IT vendor in Section 5.2.

Altinn followed up on this time-out issue. In the synchronization meeting on 12 March, Altinn said they wanted Tax *"to recreate the test where we get time-out against the ID portal, to see that it is not our tool [there is something wrong with]."* Tax thought the time-outs came because of the three-second global load balancer policy delays, as mentioned in Section 6.1. DIFI said on 12 March: *"We are having trouble figuring out the root cause of the errors we got."* DIFI discussed running a new test to go deeper into this in the same meeting but also said they had other things to do.

In the synchronization meeting on 19 March, Altinn asked DIFI again if there was something new on the time-outs in the ID portal and got a *"no"* in response. On 25 March, Altinn escalated this topic to the steering team and asked in an email about the time-outs in the ID portal tests. DIFI responded, *"We have not found a satisfactory explanation."*

6.2.3. Neglected performance requirements

The simplification in the previous section failed to identify two types of unserved users: those who visit a poster and those who only get a time-out. Consequently, explicit requirements for these two types of unserved users in the tax report SoS were missing. Nevertheless, unserved users were an underlying theme in 2015 and 2016, as Sections 3.6 and 3.7 outlined. Altinn discussed unserved users in the synchronization meeting on 5 February: *"What is important about it [Tax's decline poster] is that it can be a good tool if we do not want to send 500 [logins per second] towards the ID portal on the opening day. We must test what happens if the ID portal handles only 100 [logins] per second."* Altinn repeated this on 26 February: *"The target number should be 1,000 per second. Although* *we have now tested that [the] Tax [front end] manages 2,000, we have had few [tests] with so many users."* Table 3 in Section 5.3 shows a seven-minute average of 735 attempts a second, much higher than the expected ID portal capacity of 83 logged-in users per second.

Altinn did not convince the other partners of the importance of strict performance requirements for unserved users. Testing with many unserved users was part of the test plan but not a priority. These unserved users implicitly visited a poster. Formulating performance requirements for unserved users who got a time-out touch on the frame problem as outlined in Section 2.3.

6.2.4. Extra login verifications

The consequences of the extra login verification described in Section 4.3 were neglected because of a simplification that took a long time to uncover. People on the technical team insisted that the effect of this lightweight login verification was negligible. Intuitively, this holds for CPU consumption. However, the impact on primary memory and the number of connections in the ID portal might resemble that of a standard login. This is important as the shortage of primary memory and connections for the ID portal contributed to the outage.

This extra verification resulted from Tax's whitelist solution, indirectly due to the new Sirius pilot. The consequences of this extra login verification were neither investigated during the tax return system preparations nor during the retrospectives. As explained in Section 5.2, this extra verification was neither counted as a login nor a login attempt to the ID portal.

6.2.5. Simplified view of potential bottlenecks

Tax reports from 2011 to 2018 (see Section 3) investigated the components with saturation afterward, but they did not study the components that *almost* reached saturation or had other severe performance issues. However, with slightly different circumstances, components close to saturation could become the primary bottleneck. This was also the case in the 2019 tax return system. BankID had an outage between 12:24 and 12:28 on 4 April. Moreover, as explained in Section 6.4.3, the high utilization of Altinn's front-end server was part of the reason for the swift reaction from Altinn after the problems in the ID portal on the opening night. The utilization of the Altinn front-end servers was about 75 %, while they were tested at only 50–60 %.

6.3. Sensitivity to operations

In HROs, heedful interrelation signals "sensitivity to operations." Competition between the different partners in a cross-organizational system may threaten heedful interrelation. This section examines examples of heedful interrelation and its counterpart, competition, from different angles.

6.3.1. Different capabilities and risks for the three partners

The capabilities of the three principal partners differed. The new and unproven Sirius pilot from Tax had considerable risks, and its capacity was unknown. Nevertheless, as we have seen, Sirius was also a success.

Altinn had a polished solution. In the first status meeting on 8 January, Altinn reported that they could handle 600 users per second: *"We have significant margins, so it should not be up to us."* This is consistent with what Altinn said during the 25 April retrospective: *"We in Altinn had nothing that needed to be dealt with."*

The expected bottleneck was the two BankID authentication systems employed by the ID portal, (ordinary) BankID and Mobile BankID. This expectation came from the history of earlier tax report openings, especially in the previous years, 2017 and 2018, as explained in Sections 3.8 and 3.9. BankID was also one of the first to get the blame on the opening day, as elaborated in Section 6.4.3. It may have contributed to the problem, but as we have seen, the main problem was in the ID portal.

6.3.2. Different risk appetites for the three partners

Differences in appetite for risk between Tax and Altinn were already evident in the first status meeting on 8 January 2019 when Tax announced that they would not test the new Sirius pilot in production, as described in Section 6.1. During the 25 April retrospective, Altinn elaborated: *"We missed that the overall user [experience] had more priority than validation of the [Sirius] pilot [from Tax], but that was not how decisions were made."* It helped that the Sirius pilot could be turned off in case of problems, and its complete workload could be redirected to Altinn.

The head of the technical team (and the second author of this paper) described handling risks in December 2019: *"The three parties handle their own risk, and then we point to the others' risk. If we feel that the others are not doing anything about their risk, we escalate this [to the steering team]."* One example of escalation was handling time-outs in the ID portal described in Section 6.2.2, in which DIFI did not give sufficient priority.

Another escalation occurred on 1 March when the steering team described the Sirius pilot as *"a new, unproven system."* The email concludes that the Sirius back end from Tax has a very high probability of error and that such an error will have a significant consequence. One of the reasons for this high risk was the missing production testing of the Sirius pilot. Consequently, Altinn recommended that Tax introduce the Sirius pilot more slowly on the opening day. This would reduce the Sirius workload and, therefore, the Sirius pilot's validation. However, Tax insisted on running the Sirius pilot as planned.

The difference between Altinn and Tax in appetite for risk was also illustrated by what Tax said on 26 March, shortly before the opening day: *"We are happy with the last week. We improved our scripts and got to run the tests we wanted with the answers we wanted. We are calmer than we have been for a long time and hope to transfer some of that calm to you".* Altinn replied: *"We are worried that there is such a short time left, and we still have not landed everything. It is also a completely new solution that will interfere with us".* The comment from Tax was: *"Now you can feel how it has been for us when you have been in the driver's seat, and we have been wondering if you have control."* This refers to previous years described in Section 3 when only Altinn had a back end.

Tax's production infrastructure was not available until immediately before the opening day, as Tax moved infrastructure from its test environment to its production environment right before production. Removing infrastructure from the test environment makes it hard to reproduce potential problems. There were also changes in global logout and the smart button (described in Section 6.1) from Tax after 26 March, close to production.

6.3.3. Three partners with slightly diverging objectives

The interrelation between the constituent systems was heedful in many ways. The weekly synchronization meetings and the joint testing contributed to a shared view. Asking other partners for help was easy. They openly discussed issues and reached agreements. However, they had slightly diverging objectives.

The head of the technical team described the division of responsibility in the 2019 tax return system in December 2019: *"There are three directors and three communications managers –one in each agency. Below them, you have three working teams [one in each agency] trying to work together. Below that, you have a technical team divided into three agencies."* He also said: *"At the end of the day, you must be accountable to your own boss."*

This division of responsibility is challenging when several organizations participate in one system. During the retrospective on 25 April, Altinn said: *"We should have been run as a joint project, not as three cooperating partners. Next year, maybe Tax should be in the driver's seat, and we can participate."* Tax replied: *"We have thought the same thing… The system owner here is Tax. Then, it can be a more holistic experience, not three [different] organizations. No one aims to log on to either the Tax front end or the ID portal. They want to see their tax return."* Here, they discussed a more holistic SoS management.

In 2019, the principal partners had three individual command centers communicating via a video conference. As detailed in Section 6.4.2, Altinn only got some internal information from Tax but missed information about the ID portal.

The three principal partners had several subcontractors. Differences in incentives between the subcontractors were partly shaped by their contracts. For example, Altinn's development subcontractor had a contract that gave them penalties if Altinn went down and if it was Altinn's fault. Spots had economic consequences for the partner. Since Altinn's development subcontractor was responsible for the performance, it favored changes that improved the solution's performance.

On the other hand, Altinn's hardware subcontractor was only responsible for the solution's uptime, not its performance. Consequently, this subcontractor did not like changes in the solution since they may reduce uptime. This is a classic dilemma between the traditionally dynamic development and conservative operations departments.

During testing on 26 March (i.e., very late during testing), the technical team discovered that the partner with the highest load got the most users through the ID portal. This meant that the fraction of Altinn and Tax's throughput in the ID portal was proportional to the fraction of Altinn and Tax's arrival rate. This put heedful interrelation under stress. Altinn and Tax wanted to protect their reputations as providers of responsive public systems with sufficient capacity.

6.3.4. Shared view of performance testing

A shared view of the status of the performance testing would help in heedful interrelation. On 29 January, the technical team decided to keep an overview of all the tests. This was stated again on 26 February. However, during the 25 April retrospective, they discussed whether the problem causing the outage had occurred during testing. The most experienced performance tester in Altinn said: *"In Altinn's tests, we were never at a steady rate of 500 [arrivals per second] without errors. However, we had so many tests that I may remember incorrectly."*

The head of the technical team (and the second author) said in December 2019: *"I picked out the test plan for 2019… This is just a milestone plan. There are no details… Nowhere is it summarized what we tested, how hard we tested, and what we achieved?"* The first author described what happened during each test to the technical team but did not present this in an overall way. In December 2019, the head of the technical team said: *"I think that's the essence: We're too bad at picking up the essence. This applies to what tests have been done, how they went, and their vulnerabilities. Then you can go for the most important things."*

6.4. Commitment to resilience

According to the definition of resilience in HRO, functions should be maintained even under internal and external changes and degrade gracefully when required (Weick and Sutcliffe, 2015). This section will evaluate the resilience of the technical team in the tax return system from different perspectives.

6.4.1. Reflections from the 25 April retrospective

The 25 April retrospective details several examples of inferior commitment to resilience. The first example touches on not assuming each day will be bad and acting accordingly. In this retrospective, Tax said: *"For almost an hour, our front end had between 400 and 600 [arrivals or login requests] per second. If Altinn had the same pressure, there would have been much higher pressure on the solutions than we anticipated."* Altinn responded: *"I remind you what I said. I said we had over 1,000 [arrivals] a second [in previous tax reports, before 2019]."* This is discussed earlier in Section 6.2.3. Tax responded: *"This was a burst, but here it was for almost an hour."*

Lack of drilling in advance meant many unserved users surprised the technical team. Tax said: *"We did not train properly in this scenario of people not getting in. That's when you get the huge workload."* Altinn replied: *"We had no way to flush the users when they had already received the cookie. You keep the cookie if you do not close the browser."* Altinn again: *"As long as the browsers are up, it is theoretically impossible to eliminate them. At least if it is cookie-based."*

Lack of drilling in advance also meant it took too much time for both Altinn and Tax to reduce the load. Altinn commented at the retrospective: *"We had not drilled enough between us. We should not have throttled marginally when the ID portal went down but stopped immediately."*

6.4.2. Information available in Altinn's command center

Commitment to resilience requires a capability to learn and act. However, this capability was hampered by incomplete information about the constituent systems. While some information was available, other information was missing. On the opening day, three teams sat in three different command centers and communicated via video conference. Each team represented one principal partner, Altinn, Tax, and DIFI. A complicating factor was that the various teams had different pictures of the health of the systems in the tax report SoS. Altinn had the following information from Google Analytics in its command center:

- The number of active users for its own and Tax's web pages and the fraction of origin for these users: mobile, tablet, or computer.
- The number of page views per minute for its own and Tax's web pages.

This information was shown on two monitors – one for Altinn's information and the other for Tax's information. Moreover, Altinn showed the utilization of its most critical CPUs for its systems. Information from the ID portal was not available for the Altinn personnel.

More information was required for proper communication between the constituent systems. Internal information in other constituent systems should have been visible, e.g., the number of active users and utilizations in critical components. To track *response* times in the complete SoS, Altinn personnel repeatedly performed manual logins. This end-to-end response time metric could have been automatically collected and shown.

6.4.3. How committed to resilience were the cooperating organizations?

To describe technical teams' commitment to resilience, this section chronologically details observations, theories, and actions in the critical period between 22:48 on 3 April and 00:19 on 4 April. The different phases of the outage described in Table 3 in Section 5.3 are combined with statements from the coordination meeting during the opening.

In phase 3 in Table 3, at 22:48, DIFI reported problems with Mobile BankID. BankID immediately confirmed that one of two major Norwegian Telcos had trouble, and consequently, DIFI said, *"We're taking action."* The action meant banning users from this Telco access to prevent these problems from spreading to other parts of the solution. Such spreading was fresh in their minds from 2017 to 2018, as outlined in Sections 3.8 and 3.9.

Altinn asked at 22:50: *"How is the Sirius pilot? How many users have logged in?* with the response from Tax: *"2400 have visited Sirius."* This could be interpreted as a sign of some success. One minute later, at 22:51, Altinn said, *"We must check the batch to see if we are online,"* with the response from its development subcontractor: *"The traffic into Altinn is falling; is there anything stopping it?"* Also, at 22:51, Altinn asks: *"What about the ID portal?"* DIFI responded, *"There is reason to believe this is a consequence of the Mobile BankID error."* At 22:52, Tax confirms, *"The ID portal is down."* Also, the ID portal itself agrees that it is down at 22:53.

The three cooperating organizations spent at least the time from 22:49 until 22:52 to establish that the ID portal was down. With proper information about all constituent systems (see discussion in Section 6.4.2), this information should have been immediately available for all three cooperating organizations.

At 22.53, Altinn asked its subcontractor to lower the waiting time poster because the utilization on its front-end servers was too high. Altinn initially wanted to put it to 500 users per second for each of its two front ends, but DIFI asked them to lower it further. Therefore, at 22:54, Altinn asked the subcontractor to set the waiting time poster to 200 users per second for each of the two front ends. At 22.56, Altinn's subcontractor acknowledged that this was done, seven minutes after the outage.

When we know that the expected capacity for the ID portal was 83 requests per second (5,000 per minute), the adjustment of the waiting time poster was inadequate. It is also interesting that Altinn reacted because of problems in its front-end server and not in the ID portal.

In the meantime, at 22:54, Altinn asks, *"Can Tax also use a poster?"* i. e., limit the traffic it sends to the ID portal. At 23:01, Tax reports that *"there is now a banner on skatteetaten.no [Tax's website]."* First, at 23:22, Tax announced that it would adjust its decline poster so that 50 % of the users were declined, and at 23:28, it published that it had done this.

A poster has three meanings: a message on your web page, a waiting time poster, or a decline poster. Tax managed quite quickly to publish a message on its web page. Nevertheless, it took Tax 39 minutes to utilize its decline poster at 23:28, compared to Altinn, which used seven minutes. HROs distinguish three modes of operation: normal, up-tempo, and crisis. All business units must be in the same mode. It took some time for Altinn to change to crisis mode, but it took considerably longer for Tax.

At 22:55, DIFI thought they were victims of a Distributed Denial of System (DDoS) attack and denied access from foreign IP addresses. At 23:03, Altinn wondered whether the ID portal could reduce its decline poster, and the reply was that this had already been done. Altinn's development subcontractor had a clever remark at 23:04: *"Those who had come to Altinn before this happened have already received a cookie and live on anyway. The waiting time poster does not stop them. Many users are hammering on the ID portal."* Altinn's subcontractor elaborates: *"Most people do not close their browser automatically. I do not think it helps so much to turn down our site; I think many live on there anyway."*

All the unserved users continued to exhaust the ID portal, resulting in a new outage at 23:19 after the ID portal had nearly recovered. At 23:29, DIFI asks both Altinn and Tax to stop sending traffic entirely and acknowledges at 23:33 that there is no traffic from Tax but that it still receives traffic from Altinn. In line with earlier insights, Altinn's subcontractor responds, *"Some users of Altinn have a cookie that we cannot do anything with."* Altinn discusses how long a cookie lives and concludes that it lasts 30 minutes or until the end-user closes the browser.

Like the many attempts from Tax after 22:37, the vast number of attempts from Altinn after 23:30 points to a lack of testing with enormous numbers of unserved users. Had such testing been done, causes like misconfigurations could have been sorted out before the Big Bang.

When Altinn and Tax decided on throttling, they did not take unserved ID portal users into account, i.e., users who had passed Altinn's waiting time (and were logged into Altinn) or Tax's decline poster (and were logged into Tax) but were not yet logged in to the ID portal. They considered only the arrival rate of *new users arriving at Altinn and Tax*.

In response to a status question from Altinn, DIFI responds at 23:42: *"We managed to remove some traffic, but we are getting 60,000 requests for connections per second."* Altinn asks: *"What generates that traffic?"* DIFI replies, "*That's what we're wondering about.*" They found out Altinn had 65,000 active sessions while Tax had 26,000. The 65,000 + 26,000 = 91,000 active sessions from Altinn and Tax generated 60,000 connection requests per second. This is understandable given the enormous number of unserved users, as explained earlier in Section 5.2 and detailed in Section 6.2.4.

At 23:54, DIFI states: *"We tried to open but went straight down."* Altinn asks if the ID portal is running out of connections on the load balancer, and DIFI responds: *"It may be the firewall that is running out of connections, not the load balancer."* Altinn asks if the ID portal could be more rigid on DDoS protection. DIFI responds: *"This is legitimate traffic,"* in contrast to *illegitimate traffic* handled by DDoS protection.

As a result of the discussion, DIFI at 00:00 emptied its firewall for all users and reduced the connection time to live (TTL) from 10 minutes to 30 seconds. Around 00:08, some Altinn personnel logged in, and Altinn asked at 00:09 if it was time to open again. At 00:10, DIFI responds: *"We already have more than enough traffic. BankID reports 80 per second on their site."* DIFI states: *"We have taken three active measures: [Traffic from] abroad is still blocked. All connections in the firewall are killed. The time to live of connections in the firewall is set extremely short."* Finally, at 00:13, DIFI reports that both Tax and Altinn can send traffic to the ID portal and adds that Tax stops traffic much better than Altinn.

The insight from 23:04 on unserved users who have passed the waiting time poster is strengthened. At 00:00, they decided to eliminate these users by considerably reducing TTL. However, since the problem was in the load balancer, this reduction of TTL in the firewall had no effect. Also, killing all connections had minimal impact. As discussed earlier, connections for active users were instantly reestablished, resulting in up to 60,000 connection requests per second. Even if an eager, unserved user is a legitimate user, such users *behave* as illegitimate users and enemies of the system.

6.5. Deference to expertise

Regardless of rank, the "deference to expertise" principle gives decision authority to the most qualified people during an emergency. Examples of successful deference to expertise are:

- Knowing how to contact the relevant experts is a prerequisite for giving them authority and the basis of contact lists with information on partners not participating in the conference meeting on the opening day. The contact list had grown during earlier experiences with previous tax reports, as illustrated in Section 3.8. Therefore, when one of the two major Norwegian Telcos got a new subcontractor, DIFI said early in 2019: *"We are working to tie them closer to us so that they are awake on opening night."*
- Runtime rules in the tax report system are more subtle examples of knowing how to contact resources. The runtime rules accumulated based on bad experiences in earlier years, for instance, the permission to kill a backup running on the opening day as described in Section 3.6 and permission to disable Mobile BankID in Section 3.9. The runtime rules were mentioned in January 2019 and focused on in a separate meeting in the middle of March 2019.

There are also examples of a lack of deference to expertise:

- Tax's decision to use the decline poster took 39 minutes, in contrast to the seven minutes Altinn used to employ the waiting time poster. Tax technical personnel spent considerable time getting consent from their high-level management participating in the same Tax command center. This was not the case for the more experienced Altinn organization.
- The technical team should have been more involved when discussing technical risks. In contrast, the steering team generally managed both organizational and technical risks. When asked if they discuss more risks in the technical team now compared to in 2019, the reply from the leader of the technical team in February 2020 was: *"As of 2020, we discuss technical risks late only when we are not able to test all parts of the solution."* Further, he added: *"We talk about what happens if a component goes down. We do not talk about how likely components will go down."* The technical team's substandard handling of technical performance risks was also a problem in 2019.

7. Organizational outage causes: discussion and recommendations

This section discusses the *organizational* causes of the outage. Section 7.1, answering RQ2, presents the organizational root causes that impeded the identification of the technical causes of the outage. In Section 7.2, we summarize minor organizational causes that impeded the identification. Addressing RQ3, we present the organizational causes obstructing the outage mitigation in Section 7.3. In response to RQ4, Section 7.3 also provides recommendations linked to the different organizational causes. The relevance of SoS theory is discussed when applicable.

The first and second authors scored organizational causes on a low, medium, high, and very high scale. We define "high" and "very high" as root organizational causes, while "low" and "medium" are minor organizational causes. These scores will be used to describe the organizational influences in this section.

7.1. Organizational root causes that impeded the discovery of the outage causes

Table 4 summarizes organizational root causes that impeded the discovery of the technical causes of the outage. The first column refers to the section in which this organizational cause is described in more detail. The organizational influence described in column two is scored in column three. In the fourth column, we relate these causes to SoS management problem numbers, as described in Section 2.1.

7.1.1. Differences in worldviews and objectives

Three of these four organizational root causes marked with "Very high" influence in the third column in Table 4 are associated with the DIFI organization. Despite receiving multiple reminders during technical synchronization meetings and even from the steering team, DIFI did not prioritize investigating the causes of the outage in the ID portal performance testing environment during Tax's testing. Additionally, missing logins on the ID portal were not thoroughly analyzed. First, in January 2020, DIFI permitted the technical team to conduct testing during the ID portal outage. Since the technical team was not allowed to test the ID portal thoroughly in 2019, DIFI did not comply with the HRO principle that "skeptics improve reliability." As shown in the fourth column in Table 4, several of the SoS management problems described in Section 2.1 are relevant here:

- Problem 1: With more holistic SoS management, DIFI could have been more eager to discover what happened in its own systems.
- Problem 2: As DIFI did not address Tax's and Altinn's concerns seriously, there were poor levels of communication and support for interactions between DIFI and the two other partners.
- Problem 3: New, flexible project management mechanisms are needed to cope with an emerging dynamic environment.
- Problem 4: DIFI's inadequate attention to these issues demonstrates incompatible worldviews and divergent objectives between it and the other two partners. DIFI also hesitated to defer part of testing their ID portal to expertise in Altinn and Tax.
- Problem 5: The interdependencies stemming from Tax's involvement in the outage within DIFI's testing environment were not considered.
- Problem 7: This SoS cannot operate using too informal approaches.

At an organizational level, incompatible worldviews and diverging objectives among the constituent systems were the root causes of the outage. The project manager, Altinn, neglected tensions among organizational objectives, reflecting inadequate holistic SoS management. Holistic SoS management would facilitate collective preoccupation with failure. Then, partners can voice dissent and share otherwise restricted information. However, each partner's autonomy may weaken, striking at the heart of SoS management problems (Vargas and Braga, 2022).

Recommendations:

- **The five HRO principles enrich the SoS concept of worldview with insights applicable to software performance during testing and operation. In a start-up meeting, address the following questions with systems engineers from each constituent organization:**
  - **Shall we be preoccupied with small, emerging failures and *"celebrate them as windows into the health of the system."* (Christianson et al., 2011)?**
  - **Shall we be reluctant to simplify, and encourage a spirit of contradiction?**
  - **Shall we be cautious about *"continuing in a course of action without reevaluation"* (Weick and Sutcliffe, 2015)?**
  - **Shall we *"assume that each day will be bad and act accordingly"* (Weick and Sutcliffe, 2015)?**
  - **Shall we encourage people to reveal information that is not widely shared?**
- **If there are discrepancies between the answers from each constituent organization, discuss whether these are rooted in incompatible worldviews and diverging objectives. Is there anything to be done about this? What could be the potential consequences of not taking these things seriously?**
- **Cross-organizational retrospectives focusing on HRO principles could gradually introduce a culture of holistic SoS management.**

These recommendations will also mitigate incompatible worldviews during operation, as later elaborated in Section 7.3.3.

7.1.2. Missing performance requirements

Altinn failed to persuade the other two partners, Tax and DIFI, of the importance of rigorous performance requirements. Historically, Altinn had experienced peaks of up to 1,000 *new* users per second. In retrospect, the technical team should have specified performance requirements for two types of *unserved, existing* users: those who visit a poster and those who encounter time-outs.

Table 4
Organizational root causes that impeded the discovery of the outage causes.

Section	Organizational cause	Influence	SoS mgm problem #
6.2.2	The ID portal probably ran into problems causing the outage during testing but did not investigate this.	Very high	1, 2, 3, 4, 5, 7
6.2.2	Time-outs in the ID portal were not analyzed.	Very high	1, 2, 3, 4, 5, 7
5.4	First, in January 2020, DIFI allowed testing with so many users that the ID portal went down, and testing of ID portal users receiving time-outs was possible.	Very high	1, 2, 3, 4, 5, 7
6.2.3 6.4.1 6.4.3	Performance requirements for two types of unserved users should be explicit: those who visit a poster and those who only get a time-out.	Very high	1, 2, 3, 6, 7, 8, 9
6.2.2	The delays in Tax's global load balancer policy may have masked finding the problem in the ID portal. The new Sirius pilot caused these delays.	High	3, 4, 5, 6
6.2.4	The lightweight login was not investigated during testing. This extra verification resulted from Tax's whitelist solution, indirectly due to the new Sirius pilot.	High	4, 5, 6
6.3.1	Asymmetrical capabilities: Altinn had a polished solution. Tax had a new solution with considerable risks. DIFI had challenges with its two BankID authentication systems.	High	4
6.3.2	Tax needed to validate its Sirius pilot with a sufficient workload on the opening day.	High	4, 5, 6
6.3.3	All three partners wanted to protect their reputations and keep their systems up.	High	1, 2, 4
6.5	The technical team should be more active in discussing technical performance risks.	High	All of them

Performance requirements for unserved users are not standard textbook material in software performance engineering. However, such requirements are crucial when any of the systems in an SoS fail. A deeper examination of past experiences, such as those in 2016, outlined in Section 3.7, could have unveiled such issues. Consequently, unserved users could have been a standard part of the test scenarios in January 2019, and the memory shortage in the DIFI's load balancers could have been fixed before the opening day. However, handling time-outs from unserved users without stopping requires updated test scripts. Such scripts were implemented before testing the ID portal in January 2020.

Lacking documentation and ownership of SoS requirements and the problem of utilizing lessons learned are typical SoS management problems (Vargas and Braga, 2022). Missing performance requirements may reflect SoS management problem 6, where the impact of constituent micro-behaviors on project performance is not considered. It may also call for new flexible mechanisms for project management to cope with emerging dynamic environments, which is problem 3. Performance requirements may further be a communication vehicle for improving the communication and support for interactions between the constituent systems in problem 2 and enabling holistic SoS management in problem 1. With explicit performance requirements, conflicts for common elements in problem 8 may be sorted out. Different terminologies in problem 9 may also be mitigated. Finally, more formalism is added using performance requirements, curing problem 7.

Recommendation:

- **Utilize insights gained from previous instances of peak workload alongside potentially revised usage patterns and operating environments to revise the performance requirements.**
  - **Have the performance requirements changed?**
  - **Have new performance requirements emerged?**

7.1.3. Differences in risk appetites and asymmetrical capabilities

The technical team perceived risks primarily in Sirius, followed by the two BankID authentication systems. Combined with asymmetrical capabilities, these variations in risks led to diverging worldviews among the three principal partners:

- As outlined in Section 3, Altinn refined its solution after several incidents in previous years. When the joint testing began in January 2019, Altinn's tested capacity surpassed 600 requests per second.
- DIFI expected challenges in its two external BankID authentication systems, which had a combined capacity of 83 requests per second. The capacity of its own system was expected to be higher than this. DIFI implicitly assumed its decline poster would work, but it used the same infrastructure as the load balancers and broke down with them.
- Tax started its Sirius pilot in 2019. Validating Sirius with a sufficient workload in 2019 prepared for 2020, when Tax took over more of the tax return system. Initially, Tax wanted Sirius to handle 15 % of the tax return workload, which was later reduced to 7–8 %.

The introduction of the Sirius pilot had several side effects. First, it caused delays in Tax's global load balancer policy, which may have masked the problem in the ID portal. Second, the additional lightweight login step was introduced to increase the load on Sirius, but it also introduced extra load on other constituent systems. Preventing the technical team from testing Sirius and the lack of production testing of Sirius could have been critical. Fortunately, thanks to Sirius's success, it was unimportant.

In the preliminary evaluation of the April 2019 tax report system in January 2020, the first author wrote: *"Risk assessment was an underlying theme throughout the work, and many risks were reduced by systematic work."* The preliminary evaluation from January 2020 continues by describing how the steering team, not the technical team, did the risk assessment. The steering team risk assessment was done around 1 March 2019, i.e., quite late. They used drawings of the architecture where different parts of the solution were marked with green, yellow, or red. However, these risk architecture drawings did not seem to be an integrated part of the technical team's work.

The implicit principle that each partner was responsible for their own risks, as detailed in Section 6.3.2, challenged their openness to problems, which is essential for heedful interrelation. As advocated by Lopes et al., 2020, in Section 2.1, holistic risk assessment is especially important for SoSs. The technical team should use risks to prioritize holistic performance testing.

Recommendations:

- **Consider performance risks holistically and not only internally for each constituent system** (Lopes et al., 2020)**. Consequently, assessing performance risks should be an integrated part of the work of technical personnel.**
- **Establish a clear link between a performance risk and the corresponding performance tests.**
- **Invest in retrospectives and scrutinize the results, e.g., potential bottlenecks revealed during high workloads may represent future risks.**
- **Understand discrepancies in risk appetite among the constituent organizations.**
- **Ensure that the process required to fulfill these objectives is lightweight, so improved risk handling does not affect overall productivity.**

7.2. Minor organizational causes that impeded the discovery of the outage causes

Table 5 summarizes minor organizational causes that impeded the discovery of the technical outage causes. In this section, we will not detail the relation to SoS management problems except for indicating the management problem by number in the fourth column in Table 5.

Given that DIFI had allowed more aggressive testing of the ID portal, the technical team should have exploited differences in test tools to gain deeper insight into the workload. Increasing think times and testing the ID portal for more than the standard 10 minutes would have facilitated the problem-solving process.

The technical team should have investigated time-outs caused by global policy for users from housing associations. Further, the technical team assumed that global logout was a light call but did not investigate this assumption. Global logout turned out to be a heavy call. However, this call was only relevant to the relatively few users inside the solution.

A better overview of all the performance tests would ensure heedful interrelation. Even the most experienced Altinn tester lost the overview, as described in Section 6.3.4. However, this was less important in our case because performance tests for unserved users were missing.

Recommendation:

- **Document the essence of what has been tested, the critical findings, and possible mitigation. Connect this essence to risks.**

An improved overview of the essence of testing will ease further discussions. The process will become less reliant on the participants' good memory and more robust if, for example, key personnel quit. There is a tension between rigor and spontaneity in heedful interrelation. The discussion in the technical synchronization meeting was oral and informal. Such discussions favor extroverted individuals with prompt reactions. A more structured process, supported by models and tools, could make it easier to exploit a broader range of competencies in the team, enabling introverts to participate in the discussion actively. Furthermore, extroverts may overlook fewer critical issues. Nonetheless, a structured process that is too heavy may constrain spontaneity.

Table 5
Minor organizational causes impeding discovering the causes of the outage.

Section	Organizational cause	Influence	SoS mgm problem #
6.2.1	The technical team should have exploited differences in test tools to gain deeper insight into the workload and, therefore, use longer tests with longer think times.	Medium	3, 4, 5
6.3.4	The technical team did not provide an overview of the performance test results.	Medium	3
6.1	Implicitly, because of Sirius, time-outs for users from housing associations were not tested further.	Low	5, 6
6.1	The technical team assumed that global logout was a light call but did not investigate this assumption. Global logout turned out to be a heavy call.	Low	6
6.1	The technical team was prohibited from testing Sirius and had to use Tax's dummy test application instead.	Low	1, 2, 4

There is no contradiction between finding the essence and the "reluctance to simplify" principle. Hardware and software modeling represent ways of handling complexity and have a long tradition in software engineering. At the heart of hardware and software modeling is focusing on some crucial aspects and forgetting about other less essential aspects, but only after a thorough understanding, without simplifying.

7.3. Organizational causes that impeded the mitigation of the outage

Table 6 summarizes earlier discussions in Section 6 about organizational causes that impede the mitigation of outages. We will discuss the organizational causes in turn.

7.3.1. Missing health metrics

As described in Section 2.4, all systems should be in the same mode. HRO theory distinguishes between three modes of operation: normal, up-tempo, and crisis. Run-time information from the other constituent partners is a prerequisite for being in the same mode. Nevertheless, this information was missing because:

1) The partners underestimated the significance of this information.
2) The partners did not know which information they needed.
3) DIFI lacked sufficient trust to share internal information with Altinn or Tax.

The first and the second explanations point to missing preparations by answering HRO questions like: *"What do we count on? How could what we count on fail?"* Missing run-time information from the other constituent systems led to poor levels of communication and support for interactions between the constituent systems, which is SoS management problem 2 in Section 2.1. It also impeded new flexible mechanisms for project management to cope with emerging dynamic environments (Problem 3). Consequently, interdependencies among the constituents were not considered (Problem 5).

The partners' distrust reflects asymmetrical capabilities and risk appetites, resulting in incompatible worldviews, as elaborated in Section 7.1.3 (Problem 4). Moreover, an SoS cannot operate using informal approaches (Problem 7). Holistic SoS management (Problem 1) requires run-time information from the other constituent partners.

Co-locating technical personnel from all constituent organizations during workload peaks would enhance information sharing. If there are too many, the most critical persons could sit in one room, with others in neighboring rooms.

Table 6
Organizational causes that impeded mitigation of the outage.

Section	Organizational cause	Influence	SoS mgm problem #
6.4.2	Missing run-time information from the other partners.	Very high	1, 2, 3, 4, 5, 7
6.1	Tax's smart button did not limit login requests from existing users.	High	5, 6
6.2.2	Users waiting for ID portal login are not counted by Altinn or Tax (only by the ID portal if this was up)	High	5
6.4.1 6.4.3	Earlier action for all three partners would have enabled the ID portal to operate closer to its maximum capacity.	High	1, 2, 3, 4, 7
6.4.3	Inadequate measures from both Altinn and Tax to stop the ID portal outage.	High	1, 2, 3, 4, 7
6.4.3 6.5	Tax top-level management delayed decision-making.	High	1, 3, 4

Recommendation:

- **Evaluate if all organizations should have a common "command center" during peak workloads.**

Given that each partner has individual command centers, health metrics present another solution:

Recommendation:

- **Make sure the health of all constituent systems is available during operation. If this is not possible, discuss why.**

Sending "HTTP 429 Too many requests" with an appropriate "retry after" duration from the system facing trouble may be a partial solution. However, revealing vulnerabilities to potentially unfriendly organizations may not be wise. Furthermore, what if the organizations receiving 429 do not care about it?

7.3.2. Tax's smart button

Tax's smart button failed to restrict login attempts from existing users. Altinn tested this solution with new users. Tax and Altinn both assumed all existing users were to be served and did not imagine unserved users. This is connected to missing performance requirements in Section 7.1.2. Consequently, waiting users asking for the ID portal to log in were not counted by Altinn or Tax. These waiting users were only counted by the ID portal if it was operational.

7.3.3. Incompatible worldviews during operation

With earlier actions and adequate measures for all three teams, the ID portal could have operated closer to its maximum capacity. Each partner was concerned about their reputation, which was tied to their system throughput. They knew that the partner with the highest load got the most users through the ID portal, putting heedful interrelation under stress. Consequently, both Altinn and Tax were slow to throttle users. Altinn was more worried about Sirius than the ID portal. Therefore, to reduce risk, Altinn wanted Sirius to start with zero users and add users in 10 % increments using the smart button. Tax did not accept this and began without throttling. Tax also insisted that all users should be first sent to Tax so that they could reach their target of 7–8 % of the users. As in 2018, Altinn could have limited its users but feared losing users to Tax.

Had both Tax and Altinn reduced their workloads substantially, quickly offloading the troubled ID portal would have been possible.

However, it took time for DIFI to properly understand the situation before instructing Altinn and Tax to throttle the workload. When DIFI finally told Altinn and Tax to limit their workload, Tax's top-level management delayed decision-making. Altinn, possessing more experience, had previously deferred authority to technical personnel. However, as elaborated in Section 6.4.3, Tax ultimately was more successful than Altinn, which had not tested setting its waiting time poster to zero.

Recommendations for handling incompatible worldviews during operation are already described in Section 7.1.1.

8. Experiences after 2019

This section briefly describes experiences with tax return systems in 2020 and 2021. The participants in the 25 April retrospective discussed the opening time of the tax return system. Spreading the load over more days had always been a better choice.[1] Despite this, the three principal organizations hesitated because of bad experiences in 2014, as outlined in Section 3.5. Moreover, all citizens should ideally get their tax returns simultaneously.

From 18 March 2020 onwards, 200,000 taxpayers were noticed daily for eight days. Based on experiences in 2014, as described in Section 3.5, Tax asked if you had received the notice. However, Tax feared that reading a newspaper or listening to a friend would qualify as a notice. You had to log in using the bottleneck resource's authentication system to determine if you received a notice. However, this was not a problem.

DIFI launched a new cloud-based queue solution in 2020 for traffic from Tax, based on recommendations in the 25 April retrospective. The test scripts in 2020 were improved. Test results were also presented better. With the improved test scripts in January 2020, it was documented that the load balancers of the ID portal had little memory, causing the outage in 2019. Therefore, additional memory and CPU were purchased before the 2020 tax return system opened. In 2020, the tax support system had a streamlined project organization with Tax clearly in charge. In summary, spreading the load, a cloud-based queue solution, more ID portal load balancers resources, and a streamlined project organization gave a smoother tax report system in 2020.

In 2021, DIFI implemented a separate Altinn queue in addition to the Tax queue from 2020. This new queue further improved the handling of the tax return system in 2021. The 2021 tax return system worked fine, except when the queue reached 70,000 users. The cloud-based auto-scaled queue had been scaling up for several hours, resulting in high compute resource utilization. Tax decided too many users were in the queue and attempted to redirect them to a decline poster. However, when all these users were redirected to the auto-scaled cloud-based front-end solution containing the decline poster, that cloud system broke down. They tested removing users from the queue but did not test redirecting these users to another system. The cloud system treated all these users as unique, and autoscaling was not fast enough to mitigate the resulting tsunami. The following year, 2022, this was finetuned and tested, and they delivered the same cached decline poster to all users.

9. Factors affecting validity

The following discusses factors affecting validity, organized in accordance with Runeson and Höst's (2009) recommendations for case studies within software engineering.

9.1. Construct validity

Construct validity reflects how well the studied operational measures represent what the researcher set out to study (Runeson and Höst, 2009). Section 1 introduces the five research questions we set out to study. To study the technical issues in RQ1, we used the operational measures described in Table 1 in Section 4.4.3. All three partners measured their part of the tax report system. However, these logs were only available after the opening day and not in real time. All log data was directly measured, except for unique attempts where the number of users attempting to log into the ID portal was filtered for unique IP addresses. Section 5.2 explains how Tax had trouble filtering for unique IP addresses. Hence, unique attempts from Altinn were multiplied by a factor representing the fraction of ID portal logins from Tax to estimate the total unique attempts. Section 6.3.3 describes how the ID portal's throughput fraction was proportional to the fraction of the arrival rate for Altinn and Tax. However, Section 5.2 describes how the number of unique attempts is too conservative since some apartment residents share IP addresses. Consequently, there are more unique attempts than this number tells.

After the fact, we constructed a timeline based on measurement log data supplemented with observations from the conference calls on the opening day. The timeline was built as a cooperation between the first and second authors. Since the second author was the head of the technical team, this timeline reflects the understanding of a critical actor in this incident. Other stakeholders have also reviewed this timeline.

We found the theoretical lens of HRO theory helpful in analyzing the organizational causes in RQ2 and RQ3, contributing to the recommendations in RQ4. The coding of the data material was deductive and based on the five HRO principles. However, as discussed in Section 4.5, these HRO principles overlap. Section 4.5 discussed the relationship between the "preoccupation with failure" principle and the "reluctance to simplify" principle. This is also discussed in Section 6.1. Moreover, the lack of insight into the parts of the system operated by other organizations is described with the HRO "sensitivity to operations" principle in Section 6.3.3 and the HRO "commitment to resilience" principle in Section 6.4.2. Cantu et al. (2020) note that the five HRO principles *"can be ambiguous and difficult to translate into practice."*

9.2. Internal validity

Internal validity concerns causal relations (Runeson and Höst, 2009): *"When the researcher is investigating whether one factor affects an investigated factor there is a risk that the investigated factor is also affected by a third factor."* Understanding causation is challenging, as illustrated by the theories of root causes of the performance incident in Section 5.4. However, the final and fourth root cause theories of the outage were confirmed by measurements, so we trust this fourth theory. This case study is concerned with two causal relations:

- Understanding what technically caused the outage on the opening day.
- Understanding the organizational causes behind the outage using HRO theory.

In this case study, longitudinal involvement supports internal validity. As described in Section 4.1, we started working with the tax report opening in 2016 and have an overview of the tax report opening in 2020 and 2021, as outlined in Section 8. The three types of data sources introduced in Section 4.3 provide triangulation in data collection:

- transcribed observations of meetings
- written communication in emails and Slack, and
- measurement logs from the technical system on the opening day, 3–4 April 2019.

We studied the technical team meetings to understand causation during cooperation and coordination among the partners. Transcribed observations of meetings and written communication in emails and Slack gave us a thorough understanding of the technical discussions and the tests and measures performed.

[1] Spread-the-load is one of the principles of performance-oriented design in Smith and Williams, 2002.

Apart from Altinn, we were not closely involved with other organizations' actors and the steering team, due to time restrictions and access concerns. Thus, we may be unaware of influences and effects on the management level of other organizations and in the steering team. This is partly compensated by the key role both Altinn and the second author had in the 2019 tax report system.

Finally, the technical team may not share a full version because of a lack of trust among the participating partners or missing trust in handling the transcribed material in this research. We informed the participants how to handle research data securely to build trust. An agreement with the Norwegian Agency for Shared Services in Education and Research covered our work with the research data.

9.3. External validity

External validity *"is concerned with to what extent it is possible to generalize the findings, and to what extent the findings are of interest to other people outside the investigated case"* (Runeson and Höst, 2009). We have aimed to provide a thorough description of the case to help the reader evaluate the relevance of the findings for other contexts. Using theory is one strategy to support analytic generalization in single case studies (Yin, 2018). The findings in Section 6 apply HRO theory to our case study. Based on HRO theory, we provide recommendations in Section 7. As indicated in the fourth column in Tables 4–6, all ten problems of SoS management were relevant to this case study. The *planned* workload peak on the annual tax report opening night challenged software performance. Lessons from this planned peak can help organizations better prepare for *unexpected* peaks caused by war, pandemics, terror, or accidents.

Performance testing may seem less relevant with cloud infrastructure. Some parts of the tax report system in this case study, including Tax's front end, were already cloud-based.

However, many organizations are unable or unwilling to run critical systems on cloud infrastructure, e.g., because of legacy systems, privacy concerns, or cost. Furthermore, cloud infrastructure will not remove performance concerns caused by the Ringelmann effect, as described in Section 3.4. Our case study focuses less on the technical issues that could be partially mitigated through cloud technology and more on the organizational factors that would remain if moving to a more cloud-based system.

9.4. Reliability

Reliability *"is concerned with to what extent the data and the analysis are dependent on the specific researchers"* (Runeson and Höst, 2009). The first author did the data collection. The second author provided essential feedback on the findings. The second author was accountable for the performance in Altinn and was the head of the technical team. The close cooperation between the first and the second authors clarified the following issues in the tax report system in 2019:

- Written transcriptions and minutes from the synchronization meetings and the joint testing improved the working memory of the technical team.
- Mini retrospectives between the first and second authors after the synchronization meetings and the joint testing clarified issues for the second author.
- The first and second authors initiated more thorough retrospectives. Transcriptions from these retrospectives were shared with the technical team.
- The first and the second authors collaborated to interpret and present the log data.
- They also scored the organizational outage causes in Section 7.

The second author's active role in the technical team may influence validity. This is compensated for by defining his role as mainly covering corrections of facts on the tax report system and not the design and implementation of the case study. The third and the first author, both researchers, discussed the coding of the HRO principles and the findings. The fourth author, an experienced operational architect, checked the findings.

10. Conclusions and further work

This article shows how organizational factors caused an outage of an SoS during software performance testing. Furthermore, the lack of mitigation for organizational factors during system operation hindered the minimization of outage impact.

The most striking aspect of using HRO theory for software performance testing of a mission-critical SoS is how well the HRO principles apply. Some elements of these five HRO principles were already embedded in handling software performance during testing and operation in the tax return SoS.

Like other recent HRO theory research, our case study is not a pure HRO. Using the eight HRO criteria of Roberts and Rousseau (1989) in Section 2.3, most criteria apply, whereas criteria five on accountability and eight on multiple simultaneous outcomes fit only marginally.

We cannot confidently say that the 90-minute outage on 3 April 2019 could have been avoided if the HRO principles had been followed. Despite addressing the primary memory bottleneck in DIFI's load balancers, other potential bottlenecks existed. However, our case study indicates that handling the performance of a mission-critical SoS preparing for workload peaks shares several HRO characteristics and, consequently, can benefit from HRO theory. Using the five HRO principles, the constituent organizations can develop a collective mind fitting for holistic SoS management.

We leveraged the 2019 tax return system to address our overall research objective, as detailed by our four research questions. Section 5.4 answers RQ1 by analyzing the technical root causes of the tax return service outage. The technical root cause was a load balancer primary memory shortage caused by too many unserved users. Section 7 addresses the three remaining research questions. Answering RQ2, Section 7 uses HRO theory to present organizational root causes in an SoS that inhibit software performance testing. In line with RQ3, Section 7 also uses HRO theory to find organizational root causes that inhibit mitigating software performance issues during operation. In both cases, the organizational root cause was incompatible worldviews reflecting an inadequate holistic SoS management. Finally, Section 7 extracts recommendations addressing RQ4. The most important recommendation is to actively use the five HRO principles to develop a collective mind among the constituent organizations, bridging incompatible worldviews in handling SoS software performance during testing and operation.

Potential further work includes:

- Refine our recommendations to implement holistic software performance engineering for SoSs with additional case studies. Investigating the project management team, and not mainly the technical team, could give a deeper understanding.
- Improve the elicitation of performance requirements for SoSs. The lightweight, agile paradigm described by Brataas and Fægri, 2017 and Brataas et al., 2021, may inspire.
- Explore technological solutions to overloaded SoSs using health information, as outlined in Section 7.3.1.
- The interaction between resilience and performance was an underlying theme in the 2019 tax return SoS. Relevant questions are:
  - What happens with the other constituent systems if one constituent system goes down?
  - How long does the whole system remain up if one constituent system fails?
  - How can we modify user behavior to prevent the remaining constituent systems from going down if one constituent system goes down?

- What happens when a constituent system that was previously experiencing an outage resumes operation?
- How fast can users declined during an outage be served when the whole system works again?

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT and Claude in order to improve the language in the final revision of this paper. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

CRediT authorship contribution statement

Gunnar Brataas: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Data curation, Conceptualization. **Petter Braskerud:** Writing – review & editing, Resources, Data curation. **Inger Anne Tøndel:** Validation, Methodology, Conceptualization. **Steinar Kjærnsrød:** Visualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Petter Braskerud has worked as a technical architect for the Altinn part of the Norwegian Digitalization Agency and its predecessors from 1 August 2013 until 31 August 2022. He was responsible for the performance of the Altinn solution in the tax return service in 2019. Petter Braskerud now works for the Norwegian Armed Forces.

Both Braskerud and his former employer, the Norwegian Digitalization Agency, want a complete and objective picture of what happened in the tax return service in April 2019 so that Altinn and the rest of the partners can learn based on this example.

Braskeruds active role in the technical team may create bias and influence validity. This is compensated for by defining his role to cover only corrections of facts on the tax report service and not the design and implementation of the case study.

The other authors were not employed by any organizations involved in the tax return service in April 2019.

Acknowledgments

The research leading to these results received funding from the Research Council of Norway, grant # 309811, Capacity Guard of Public Digital Systems. We thank the Norwegian Digitalisation Agency (now responsible for Altinn and the ID portal) and the Norwegian Tax Administration for access to mission-critical information and critical stakeholders. We thank Nancy Lea Eik-Nes for input in revising the text.

Data availability

The authors do not have permission to share data.

References

Agwu, A.E., Labib, A., Hadleigh-Dunn, S., 2019. Disaster prevention through a harmonized framework for high reliability organisations. Saf. Sci. https://doi.org/10.1016/j.ssci.2018.09.005.

Bezemer, C.P., et al., 2019. How is performance addressed in DevOps?. In: International Conference on Performance Engineering. ACM. https://doi.org/10.1145/3297663.3309672.

Brataas, G., Fægri, T.E., 2017. Agile scalability requirements. In: Proc of International Conference on Performance Engineering. ACM. https://doi.org/10.1145/3030207.3030240.

Brataas, G., Herbst, N., Ivansek, S., Polutnik, J., 2017. Scalability analysis of cloud software systems. In: 2017 IEEE International Conference on Autonomic Computing (ICAC), pp. 285–292. https://doi.org/10.1109/ICAC.2017.34.

Brataas, G., Hanssen, G.K., Herbst, N., van Hoorn, A., 2020. Agile scalability engineering: the ScrumScale method. IEEE Softw. 37. https://doi.org/10.1109/MS.2019.2923184.

Brataas, G., Martini, A., Hanssen, G.K., Ræder, G., 2021. Agile elicitation of scalability requirements for open systems: a case study. J. Syst. Softw. 182, 111064. https://doi.org/10.1016/j.jss.2021.111064.

Cantu, J., Tolk, J., Fritts, S., Gharehyakheh, A., 2020. High reliability organization (HRO) systematic literature review: discovery of culture as a foundational principle. J. Contingencies Crisis Manage. 00, 1–12. https://doi.org/10.1111/1468-5973.12293.

Cantu, J., Gharehyakheh, A., Fritts, S., Tolk, J., 2021. Assessing the HRO: tools and techniques to determine the high-reliability state of an organization. Saf. Sci. 134. https://doi.org/10.1016/j.ssci.2020.105082.

Christianson, M.K., Sutcliffe, K.M., Miller, M.A., Iwashyna, T.J., 2011. Becoming a high reliability organization. Crit. Care 15. https://doi.org/10.1186/cc10360. Number.

Grabowski, M., Roberts, K.H., 2019. Reliability seeking virtual organizations: challenges for high Reliability organizations and resilience engineering. Saf. Sci. 512–522. https://doi.org/10.1016/j.ssci.2016.02.016.

Han, X., et al., 2023. A Systematic Mapping Study of Software Performance Research, Software Practice and Experience. Wiley. https://doi.org/10.1145/2851613.2851978.

Hartvigsen, T., Feiring, A., Haarberg, A.M.S., Korsveien, S., Skogan, D., 2012. Vurdering av Altinn II plattformen (in Norwegian). Rapport Nr. 2011-1239. Revisjon 1.1. Det Norske Veritas. https://kudos.dfo.no/dokument/10693/vurdering-av-altinn-ii-plattformen.

Lopes, S.d.S., et al., 2020. Risk management for system of systems: a systematic mapping study. In: IEEE International Conference on Software Architecture Companion, pp. 258–265. https://doi.org/10.1109/JSYST.2020.3018068.

Nielsen, C.B., et al., 2015. Systems of systems engineering: basic concepts, model-based techniques, and research directions. ACM Comput. Surv. 48, 18. https://doi.org/10.1145/2794381. Article.

Roberts, K.H., 1989. New challenges in organizational research: high reliability organizations. Ind. Crisis Q. 3 (2), 111–125. https://doi.org/10.1177/108602668900300200 page(s).

Roberts, K.H., Rousseau, D.M., 1989. Research in nearly failure-free, high-reliability organizations: having the bubble. IEEE Trans. Eng. Manage. 36 (2), 132–139. https://doi.org/10.1109/17.18830.

Runeson, P., Höst, M., 2009. Guidelines for conducting and reporting case study research in software engineering. Empir. Softw. Eng. 14, 131. https://doi.org/10.1007/s10664-008-9102-8.

Salovaara, A., Lyytinen, K., Penttinen, E., 2019. High reliability in digital organizing: mindlessness, the frame problem, and digital operations. MIS Q. 43 (2), 555–578. https://doi.org/10.25300/MISQ/2019/14577.

Smith, C.U., Williams, L.G., 2002. Performance Solutions: A Practical Guide to Creating Responsive, Scalable Software. Addison Wesley.

Sutcliffe, K.M., 2011. High reliability organizations (HROs). Best Pract. Res. Clin. Anaesthesiol. 25. https://doi.org/10.1016/j.bpa.2011.03.001.

Tøndel, I.A., Brataas, G., 2022. SecureScale: exploring synergies between security and scalability in software development and operation. In: Proceedings of the 2022 European Interdisciplinary Cybersecurity Conference (EICC '22). ACM, pp. 36–41. https://doi.org/10.1145/3528580.3528587.

Vargas, I.G., Braga, R.T.V., 2022. Understanding system of systems management: a systematic review and key concepts. IEEE Syst. J. 16 (1). https://doi.org/10.1109/JSYST.2020.3018068.

Weick, K.E., Sutcliffe, K.M., Obstfeld, D., 1999. Organizing for high reliability: processes of collective mindfulness. In: Sutton, R.S., Staw, B.M. (Eds.), Research in Organizational Behavior, Research in Organizational Behavior, 1. Jai Press, Stanford, pp. 81–123.

Weick, K.E., Sutcliffe, K.M., 2001. Managing the Unexpected, 1st edition. Jossey Bass.

Weick, K.E., Sutcliffe, K.M., 2007. Managing the Unexpected, 2nd edition. Jossey Bass, Wiley.

Weick, K.E., Sutcliffe, K.M., 2015. Managing the Unexpected, 3rd edition. Wiley.

Woodside, M., Franks, G., Petriu, D.C., 2007. The future of software performance engineering. Future of Software Engineering. IEEE, pp. 171–187. https://doi.org/10.1109/FOSE.2007.32.

Yin, R.K., 2018. Case Study Research and Applications: Design and Methods, 6th edition. Sage.